

Level 4

Watsonx Orchestrate

Hands-on lab guide – Using ADK with SaaS



Version 3.1

July 2026

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Introduction

About this lab

This lab aims at providing you with a high-level introduction on how to work with a watsonx Orchestrate SaaS tenant from an on prem watsonx Orchestrate Developer Edition using the watsonx Orchestrate Agent Development Toolkit (ADK).

At the end of this lab, you will be able to:

- Create an environment in your ADK connected to a SaaS instance
- Create tools, knowledge, agents and connections from the ADK inside a SaaS tenant
- Remove some tools, agents, toolkit or connection from a SaaS tenant using the ADK

Environments

This lab has been created and tested using 2 environments:

- A local ADK 2.0 environment
- A watsonx Orchestrate SaaS environment hosted on IBM Cloud

How to get support

If you encounter an issue with this lab,

- IBM employees can open an IBM Support ticket at <http://ibm.biz/support>.
- Business Partners can request help at the [Partner Plus Support](#) website.

Prerequisites

Make sure the following prerequisites are satisfied before proceeding with this lab:

- You have performed the Preparation Lab guide including the environment validation step
- You have your local ADK 2.12 or higher environment up and running (Orchestrate server started after validating the environment in the preparation guide OR after having performed another ADK related lab)

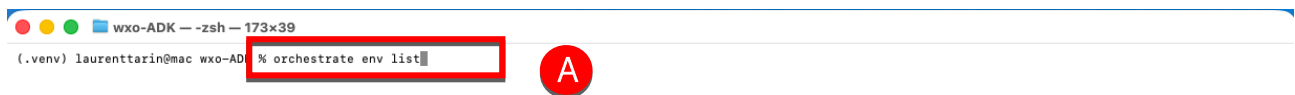
Note: If for some reason your server is not started correctly, reset your server following the [Reset your environment](#) appendix.

Create a SaaS environment in your ADK

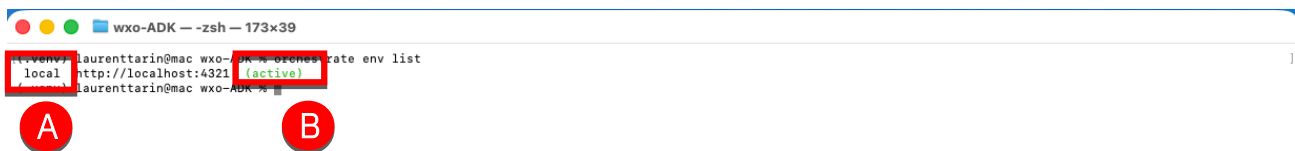
Connect to your VMWare environment

This lab requires to use the VM ware image with the ADK pre-installed you have provisioned following the Preparation guide instructions.

1. Open the ADK environment and start the watsonx Orchestrate server if not yet done. Refer to the [Reset your environment appendix](#) if your orchestrate server is not started.
2. You can see the environments you have access to from your ADK using the env command. Type 'orchestrate env list' (A).

A terminal window titled 'wxo-ADK --zsh -- 173x39'. The prompt is '(.venv) laurenttarin@mac wxo-ADK %'. The command 'orchestrate env list' is being typed and is highlighted with a red box. A red circle with the letter 'A' is next to the terminal.

3. By default, only one **local** environment is created (A). This is the default Draft environment that is only supported and available in the watsonx Orchestrate Agent Developer Toolkit. It's status is **active** (B). Currently all the orchestrate commands are applied to this environment.

A terminal window titled 'wxo-ADK --zsh -- 173x39'. The prompt is '(.venv) laurenttarin@mac wxo-ADK %'. The command 'orchestrate env list' has been executed. The output is 'local http://localhost:4321 (active)'. The output line is highlighted with a red box. A red circle with the letter 'A' is next to the prompt, and a red circle with the letter 'B' is next to the output line.

Add a new environment

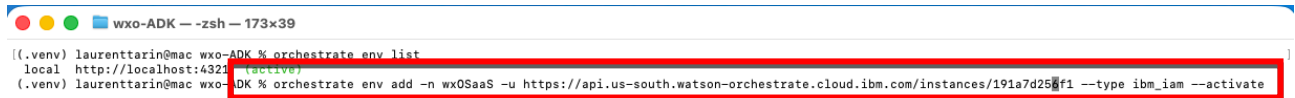
In this section, you will create a new environment that will be connected to your watsonx Orchestrate SaaS tenant you have reserved during the preparation steps. At this stage you have collected:

- Your SaaS Service URL (Item 7)
- The API key to connect the instance (Item 8)

The environment will be created using these credentials.

You will now create a new environment called “wxOSaaS” using the orchestrate env add command.

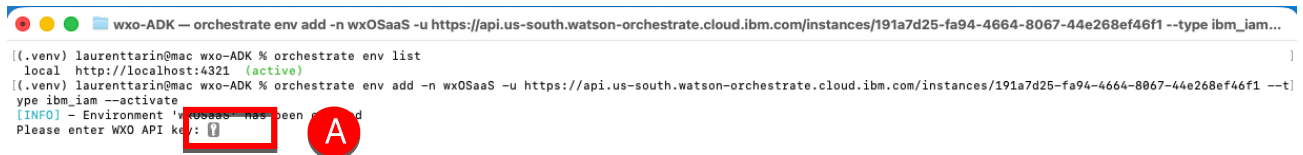
1. Type the following command replacing **<your wxo saas url>** with your own URL collected in Item 7
(A): `orchestrate env add -n wxOSaaS -u <your wxo saas url> --type ibm_iam --activate`



```

(.venv) laurenttarin@mac wxo-ADK % orchestrate env list
local http://localhost:4321 (active)
(.venv) laurenttarin@mac wxo-ADK % orchestrate env add -n wxOSaaS -u https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/191a7d25-fa94-4664-8067-44e268ef46f1 --type ibm_iam --activate
  
```

2. Copy and paste your wxO SaaS API key (Item 8) when prompted to do so (A) and immediately type the [Enter] key to submit.

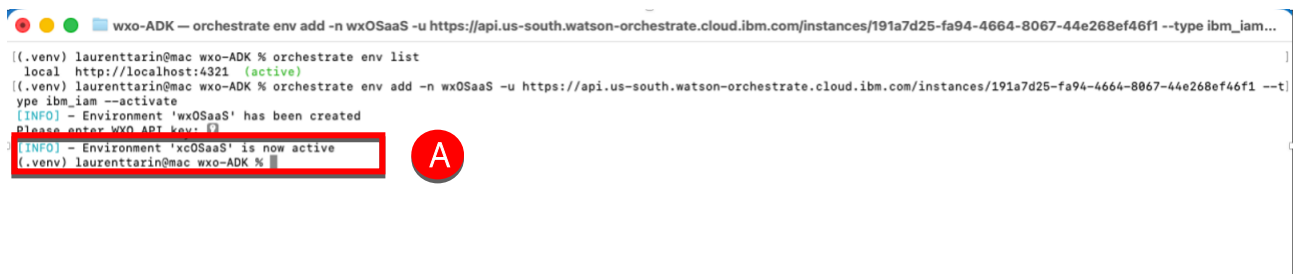


```

(.venv) laurenttarin@mac wxo-ADK % orchestrate env add -n wxOSaaS -u https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/191a7d25-fa94-4664-8067-44e268ef46f1 --type ibm_iam --activate
[INFO] - Environment 'wxOSaaS' has been created
Please enter WxO API key: 
  
```

Note: You will see no UI changes after pasting the API key. Just copy once and then hit the enter key to validate the key.

3. Once done you should see the environment creation and activation message (A).



```

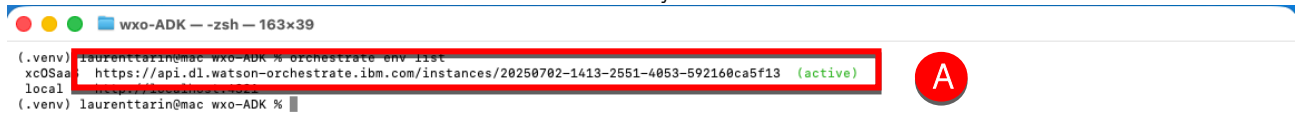
(.venv) laurenttarin@mac wxo-ADK % orchestrate env list
local http://localhost:4321 (active)
(.venv) laurenttarin@mac wxo-ADK % orchestrate env add -n wxOSaaS -u https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/191a7d25-fa94-4664-8067-44e268ef46f1 --type ibm_iam --activate
[INFO] - Environment 'wxOSaaS' has been created
Please enter WxO API key: 
[INFO] - Environment 'wxOSaaS' is now active
(.venv) laurenttarin@mac wxo-ADK % 
  
```

4. Type 'orchestrate env list' (A) to see the list of available environments on your ADK instance



```
(.venv) laurenttarin@mac wxo-ADK % orchestrate env list
```

5. You can see that your active environment is now the WXO-SaaS one (A). All the orchestrate commands are now sent to this environment from your ADK instance.

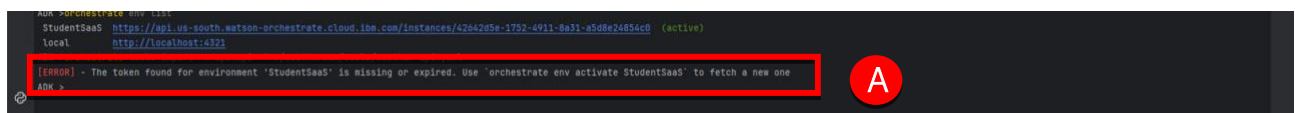


```
(.venv) laurenttarin@mac wxo-ADK % orchestrate env list
wxOSaaS https://api.dl.watson-orchestrate.ibm.com/instances/20250702-1413-2551-4053-592160ca5f13 (active)
local http://localhost:4321
(.venv) laurenttarin@mac wxo-ADK %
```

Reconnect your environment after a token timeout

During this lab it may happen that your access token expires, and the connection must be re-activated. Follow this procedure to reactive an expired environment when you see the error message.

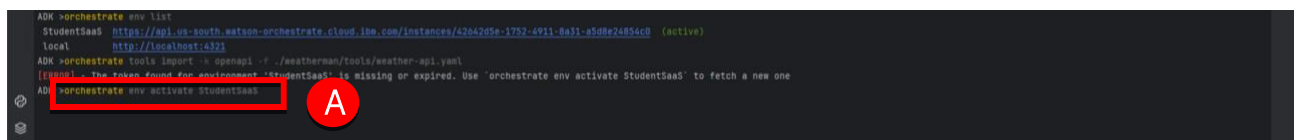
1. If your access token has timed out and you may see the following error message (A):



```
ADK >orchestrate env list
StudentSaaS https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/4204205e-1752-4911-8a31-a5d8e24854c0 (active)
local http://localhost:4321
[ERROR] - The token found for environment 'StudentSaaS' is missing or expired. Use 'orchestrate env activate StudentSaaS' to fetch a new one
ADK >
```

2. Whenever you are facing this issue just type the following command (A) to activate your environment again.

```
orchestrate env activate wxOSaaS
```



```
ADK >orchestrate env list
StudentSaaS https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/4204205e-1752-4911-8a31-a5d8e24854c0 (active)
local http://localhost:4321
ADK >orchestrate tools import -f ./weatherman/tools/weather-api.yaml
[ERROR] - The token found for environment 'StudentSaaS' is missing or expired. Use 'orchestrate env activate StudentSaaS' to fetch a new one
ADK >orchestrate env activate StudentSaaS
```

Note: this command can also be used to swap from your active environment to a new one.

3. Re-enter your SaaS **API key** (A) and immediately press enter.

```
StudentSaaS https://api.us-south.watson.ibm.com/instances/c2c7d8e-1792-d911-b611-a0d8e7c0f6dc (Active)
Local http://localhost:4321
ADK >orchestrate tools import -k openapi -f ./weatherman/tools/weather-api.yaml
[ERROR] - The token found for environment 'StudentSaaS' is missing or expired. Use 'orchestrate env activate StudentSaaS' to fetch a new one
ADK >orchestrate env activate StudentSaaS
Please enter WXG API Key:
```

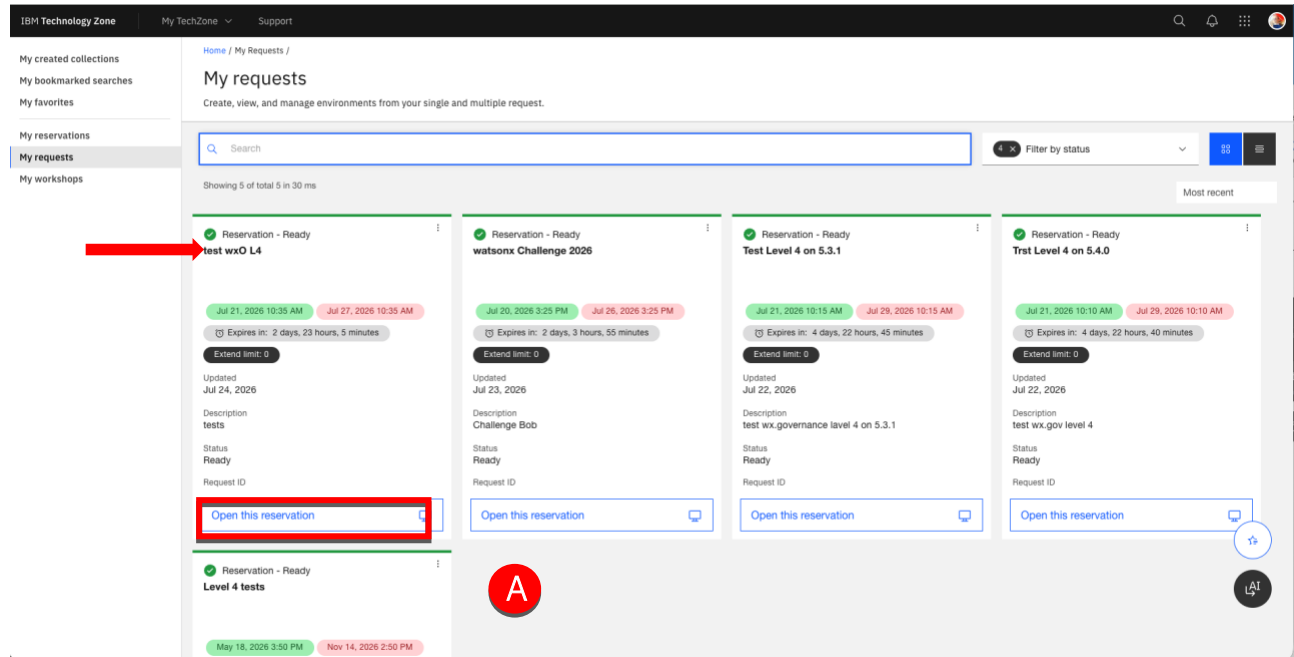
4. You should then get the active message again (A):

```
ADK > orchestrate tools import --k openapi --f ../weatherman/tools/weather-api.yaml
[WARN] - The token found for environment 'StudentSaas' is missing or expired. Use 'orchestrate env activate StudentSaas' to fetch a new one
ADK > orchestrate env activate StudentSaas
Please check the environment status
[INFO] - Environment 'StudentSaas' is now active
```

Open your SaaS environment

During this lab, you will have no specific actions to perform on the watsonx Orchestrate SaaS environment, but we will monitor the impact of the ADK commands on the SaaS instance.

1. Open your IBM TechZone reservation page <https://techzone.ibm.com/my/requests>
2. Click the **Open the environment** you have requested for watsonx Orchestrate in the preparation step (A).



3. Expand the reservation details (A).

The screenshot shows the IBM Technology Zone interface. The top navigation bar includes 'IBM Technology Zone', 'My TechZone', and 'Support'. The left sidebar lists 'My created collections', 'My bookmarked searches', 'My favorites', 'My reservations', 'My requests' (highlighted), and 'My workshops'. The main content area is titled 'test wxO L4' and shows a timeline with dates 'July 21, 2026 10:35 AM' and 'July 27, 2026 10:35 AM', and an expiration notice 'Expires in: 2 days, 23 hours, 4 minutes'. Below this is a 'Status' section with a progress bar showing stages: Requested, Pending Approval, Scheduled, Provisioning, Ready (active), and Deleted. The 'Details' section contains an 'Environments' table with columns 'Name', 'ID', and 'Status'. The table has one row: 'watsonx Orchestrate Dedicated Trial (Account Vending)' with ID '6a5f2ea62c0e4d555929e050' and status 'Ready'. A red circle with a white 'A' highlights a dropdown arrow in the 'Name' column of this row. The table also includes pagination controls: 'Items per page: 10', '1~1 of 1 items', and '1 of 1 page'. A 'Cookie Preferences' button is in the bottom right corner.

Name	ID	Status
▼ watsonx Orchestrate Dedicated Trial (Account Vending)	6a5f2ea62c0e4d555929e050	Ready

4. Open the IBM Cloud clicking the assigned URL (A). Note the user/password for the next step (B).

IBM Technology Zone My TechZone Support

My created collections
My bookmarked searches
My favorites
My reservations
My requests
My workshops

Filter table

Name	ID	Status
watsonx Orchestrator Dedicated Trial (Account Vending)	6a5f2ea62c0e4d555929e050	Ready

Collapse All

OUTPUTS

Additional Instructions

****IMPORTANT****: DO NOT USE the default IBM Cloud login. Use the IBM Cloud Login URL from this reservation. You must be logged out of IBM Cloud before authenticating, or use an Incognito/Private browser session.

Use the following Credentials to Login to IBM Cloud IDP

IBM Cloud IDP Access:	https://cloud.ibm.com/authorize/2838578d5213410fbc35fbb4cc69812d/yXDwIRkj8n				
IBM Cloud IDP User List:	<table><thead><tr><th>User name</th><th>Password</th></tr></thead><tbody><tr><td>student0@6a5f2ea62c0e4d555929e050.techzone.com</td><td>ZKtmgpBeygWyAS1r</td></tr></tbody></table>	User name	Password	student0@6a5f2ea62c0e4d555929e050.techzone.com	ZKtmgpBeygWyAS1r
User name	Password				
student0@6a5f2ea62c0e4d555929e050.techzone.com	ZKtmgpBeygWyAS1r				
IBM Cloud Service ID:	ServiceId-8f77cc45-0476-434e-bc92-1e179ebe2daa				
IBM Cloud Service API key:	L_SAagnHBH3Btzz5FnNa-6ta5Eh9U15_G1avWG5xJG8				

SaaS Link

<https://cloud.ibm.com/authorize/2838578d5213410fbc35fbb4cc69812d/yXDwIRkj8n>

5. Enter your User/password from previous step (A) and Click Sign in (B). Then follow the login instructions.

Learn. Build. Show. Share

Username

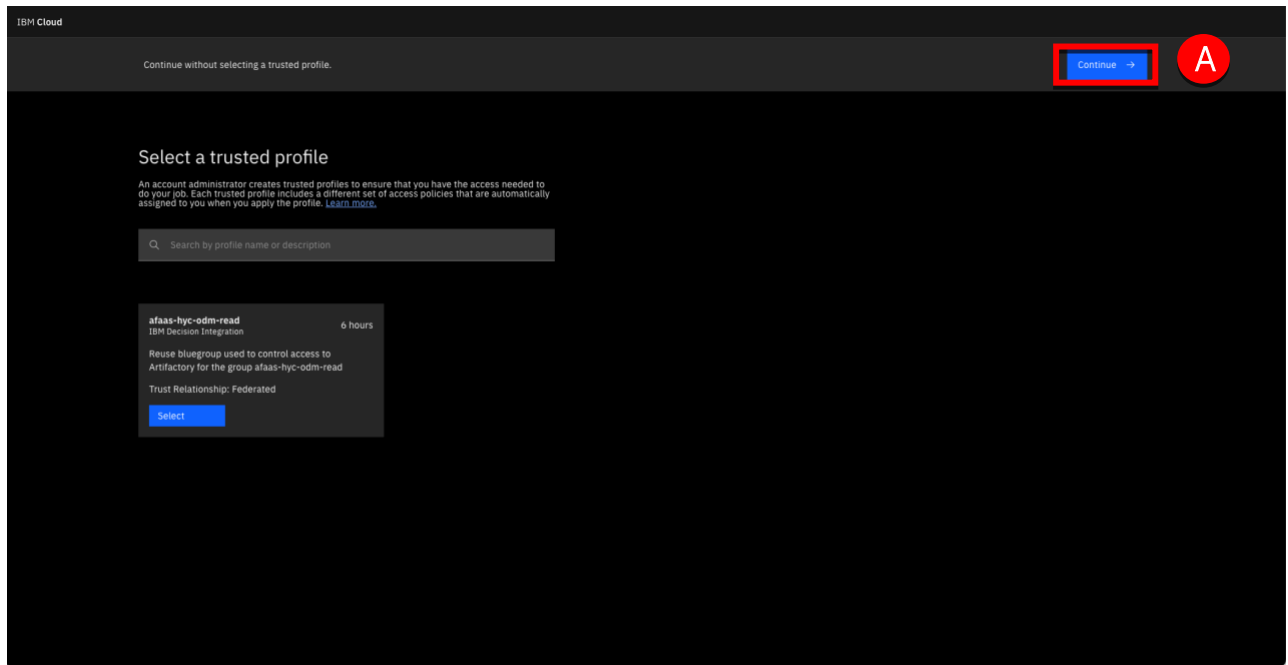
student0@6a5f2ea62c0e4d555929e050.techzone.com

Password

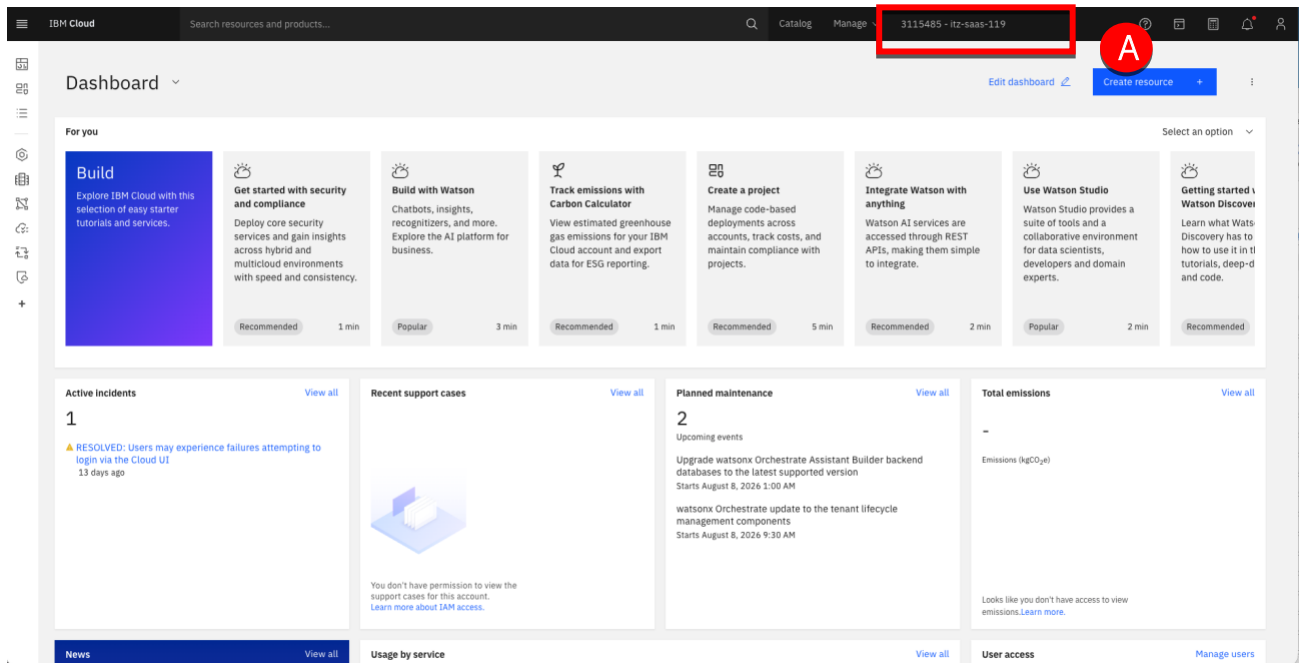
Sign in

Powered by IBM Technology Zone

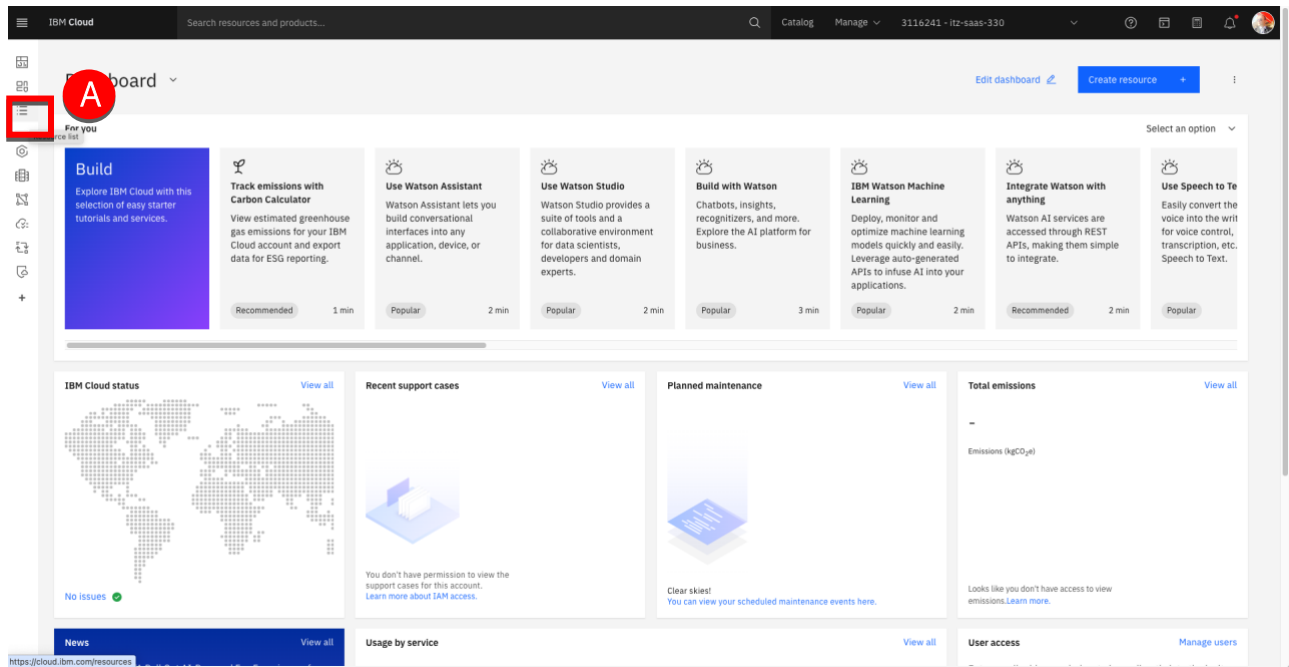
6. Click Continue (A).



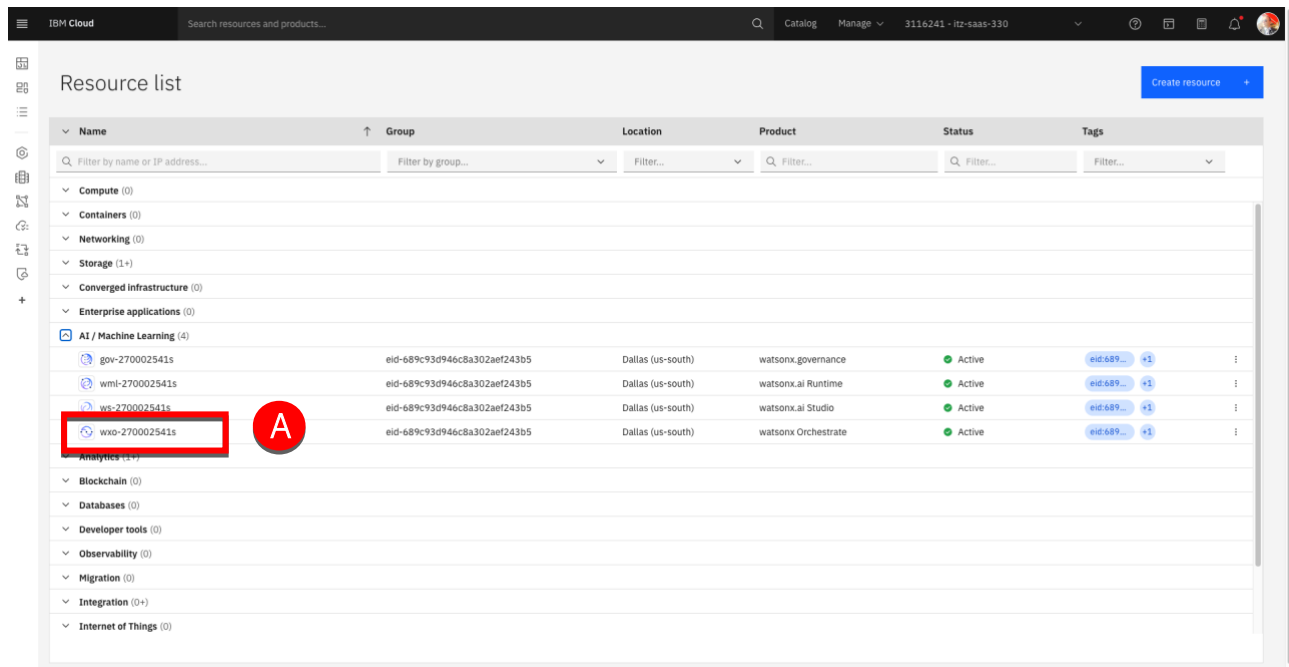
7. Note your assigned cloud account for later connections (A).



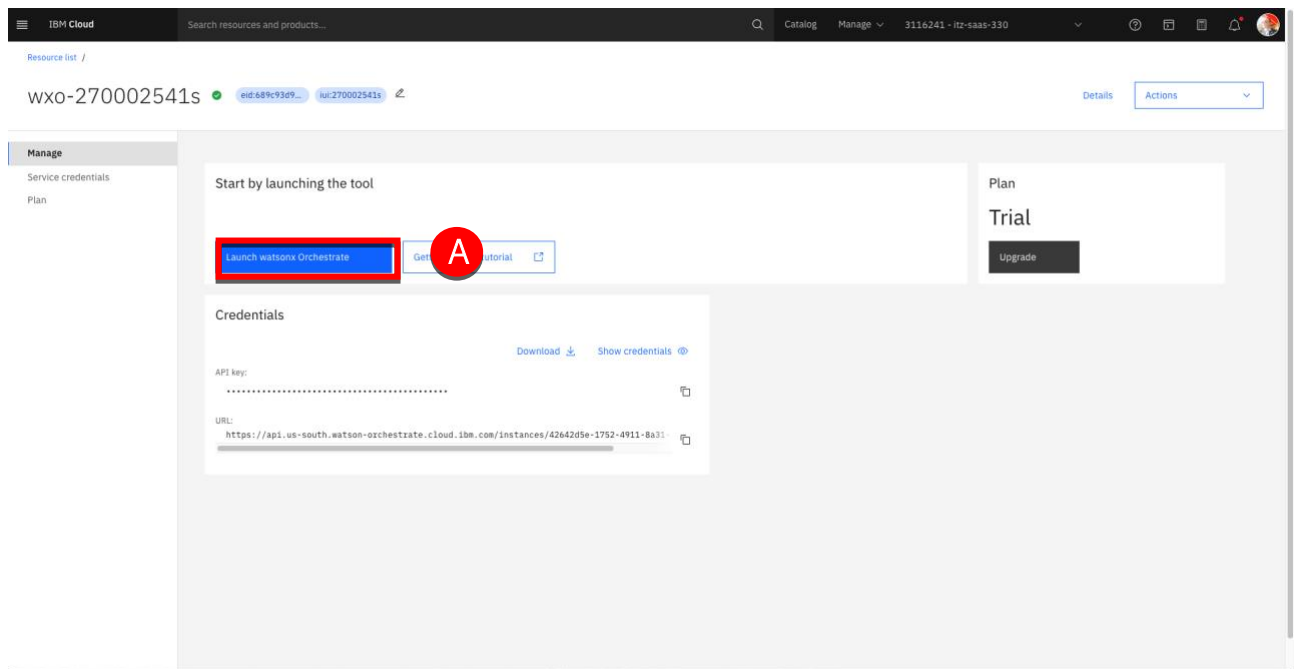
8. Click the **Resource list** icon (A).



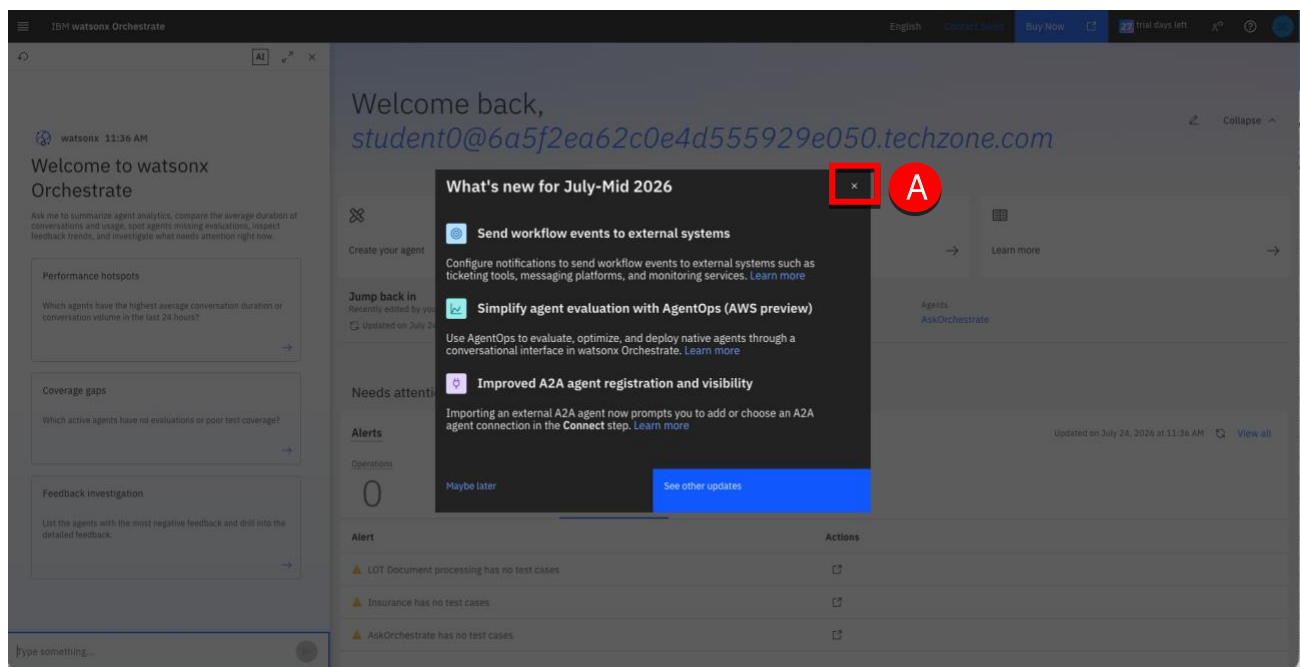
9. Expand **AI / Machine Learning** (A) then click **wxo-#####** (B).



10. Click Launch watsonx Orchestrate (A).



11. Close the welcome page (A).



12. Your watsonx Orchestrate SaaS instance is ready.

IBM watsonx Orchestrate

EnglishContact SalesBuy Nowtrial days left

watsonx 11:36 AM

Welcome to watsonx Orchestrate

Ask me to summarize agent analytics, compare the average duration of conversations and usage, spot agents missing evaluations, inspect feedback trends, and investigate what needs attention right now.

Performance hotspots

Which agents have the highest average conversation duration or conversation volume in the last 24 hours?

Coverage gaps

Which active agents have no evaluations or poor test coverage?

Feedback investigation

List the agents with the most negative feedback and drill into the detailed feedback.

Type something...

Welcome back,
student0@6a5f2ea62c0e4d555929e050.techzone.com

Create your agent

Explore agent catalog

Learn more

Jump back in

Recently edited by your team

Updated on July 24, 2026 at 11:36 AM

Agents
LOT Document processing

Agents
Insurance

Agents
AskOrchestrate

Needs attention

Alerts

Updated on July 24, 2026 at 11:36 AM

View all

Operations

Incidents

Insights

0

0

3

Alert	Actions
LOT Document processing has no test cases	
Insurance has no test cases	
AskOrchestrate has no test cases	

Manage tools in your SaaS tenant from your ADK

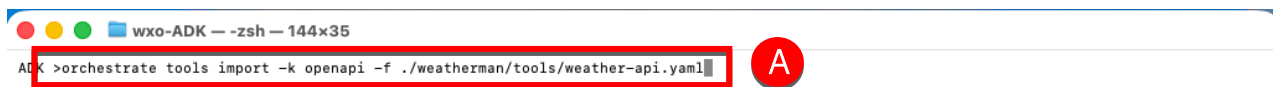
In this section you will learn how to create tools and see all tools that are deployed on the SaaS tenant from your on prem ADK.

Import a new tool

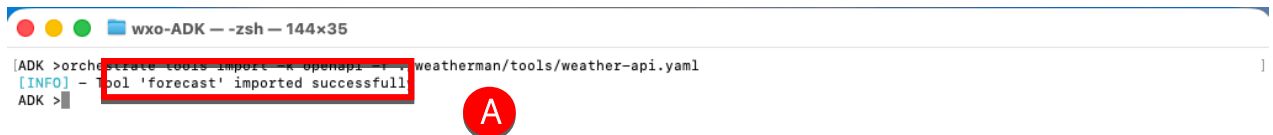
In this section you will import the 'forecast' tool used by the weatherman agent you may have already created from the ADK lab. You will use the exact same command but this time, as your active environment is the SaaS one, the tool will be created in the SaaS.

1. In your ADK environment, type the following command (A).

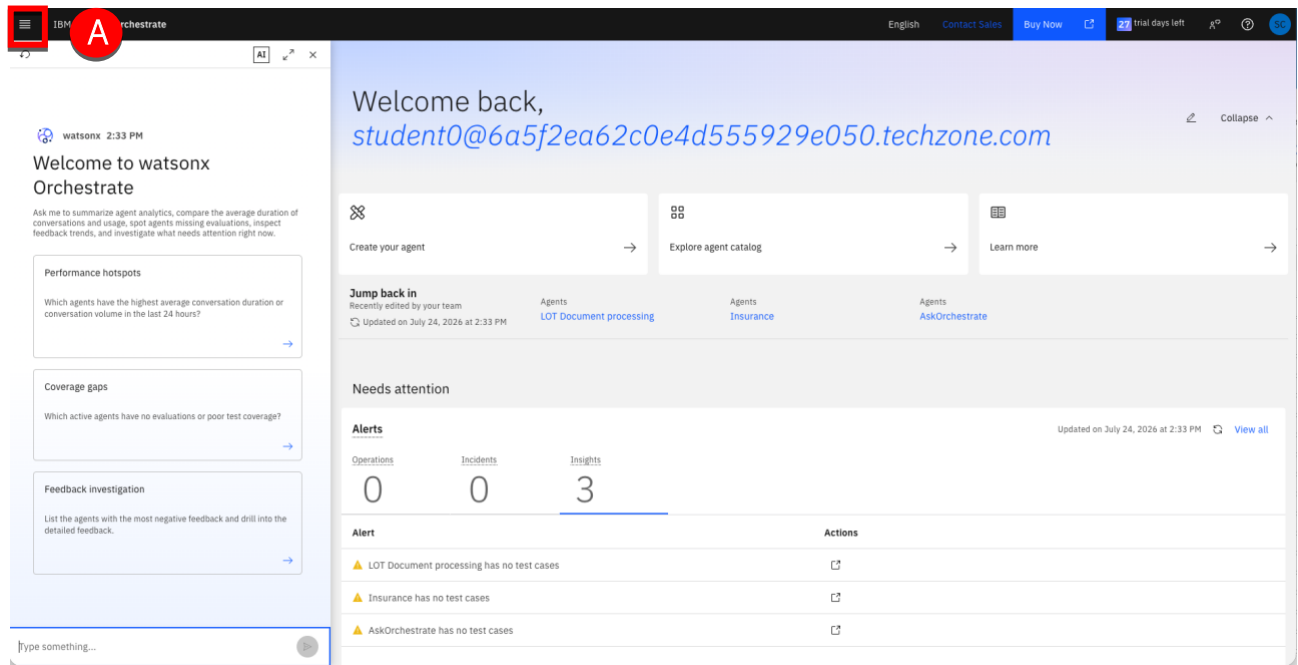
```
orchestrate tools import -k openapi -f ./weatherman/tools/weather-api.yaml
```



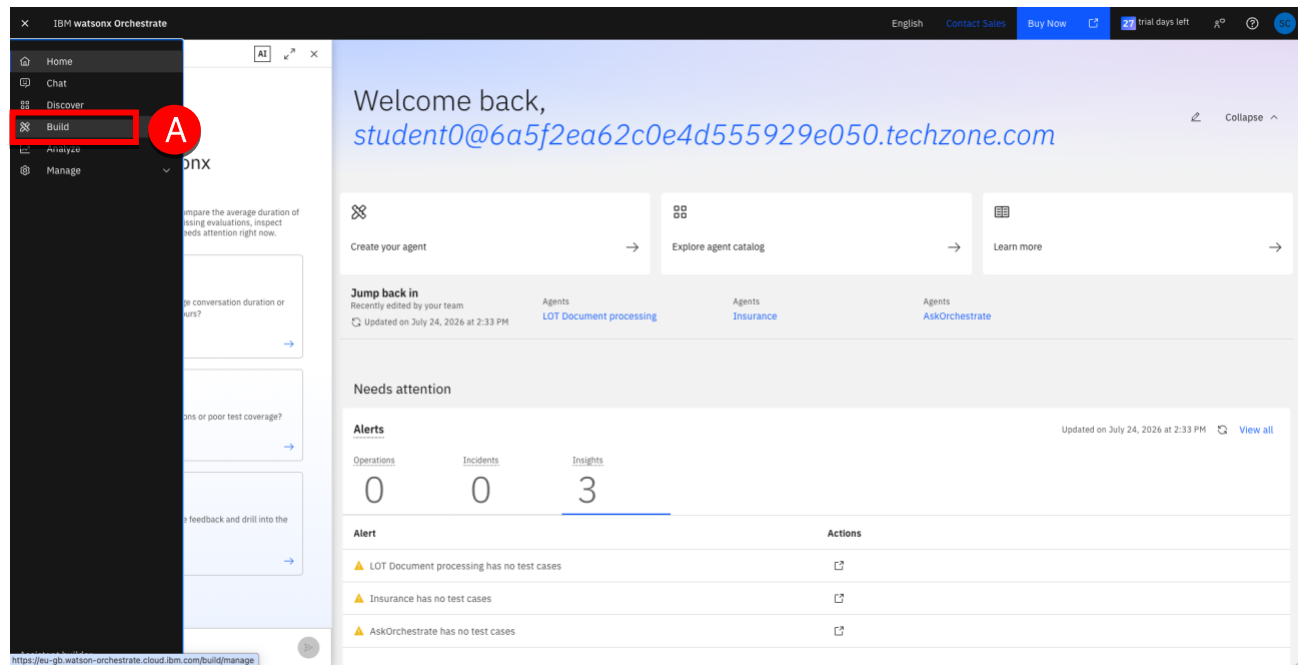
2. Wait for the tools import confirmation message (A)



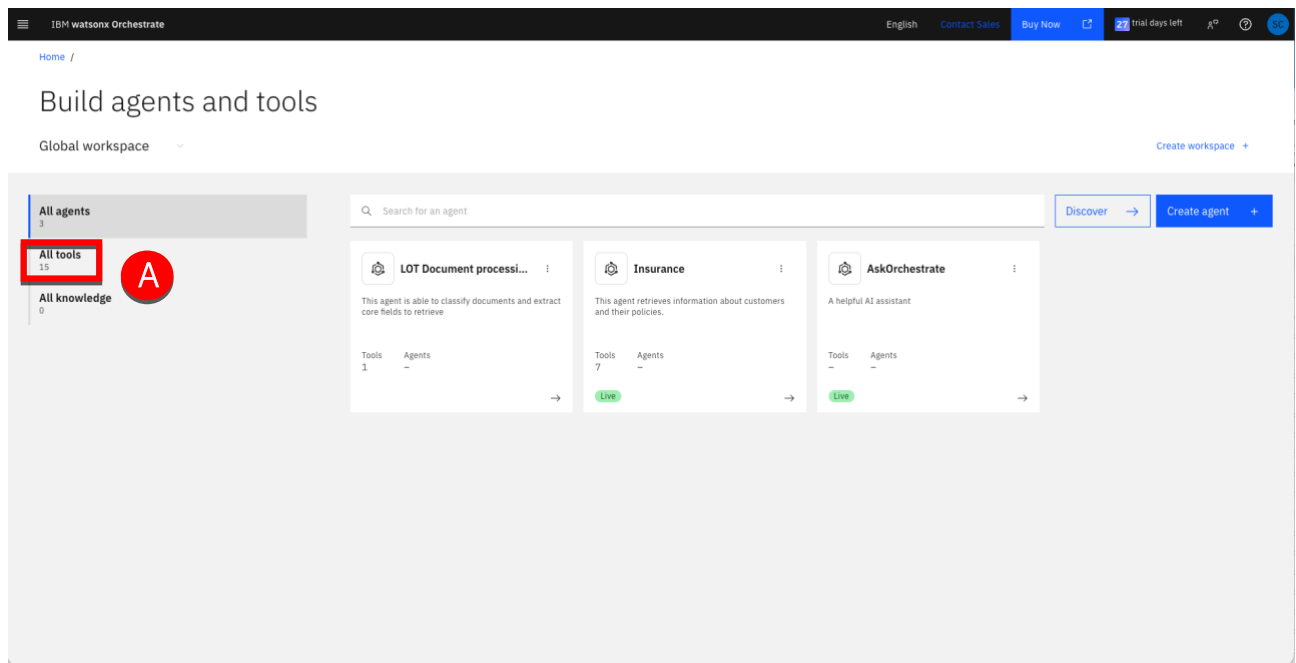
3. In your SaaS instance, click the hamburger icon menu (A).



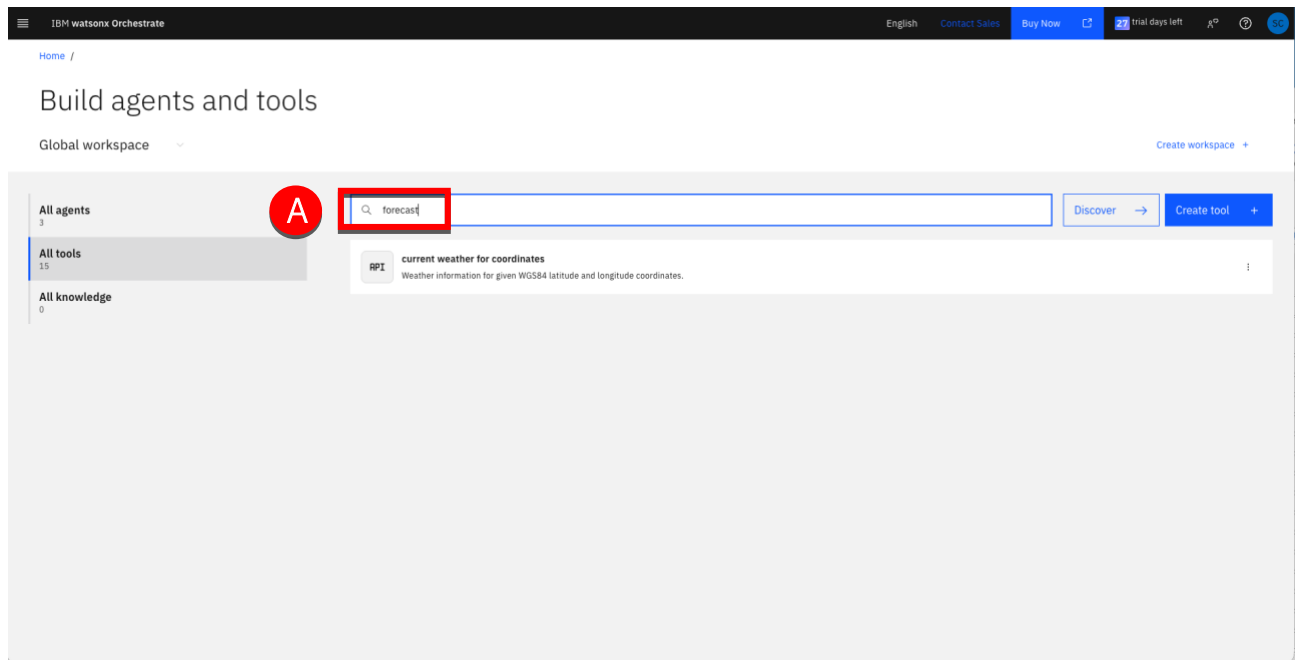
4. Click the Build menu (A).



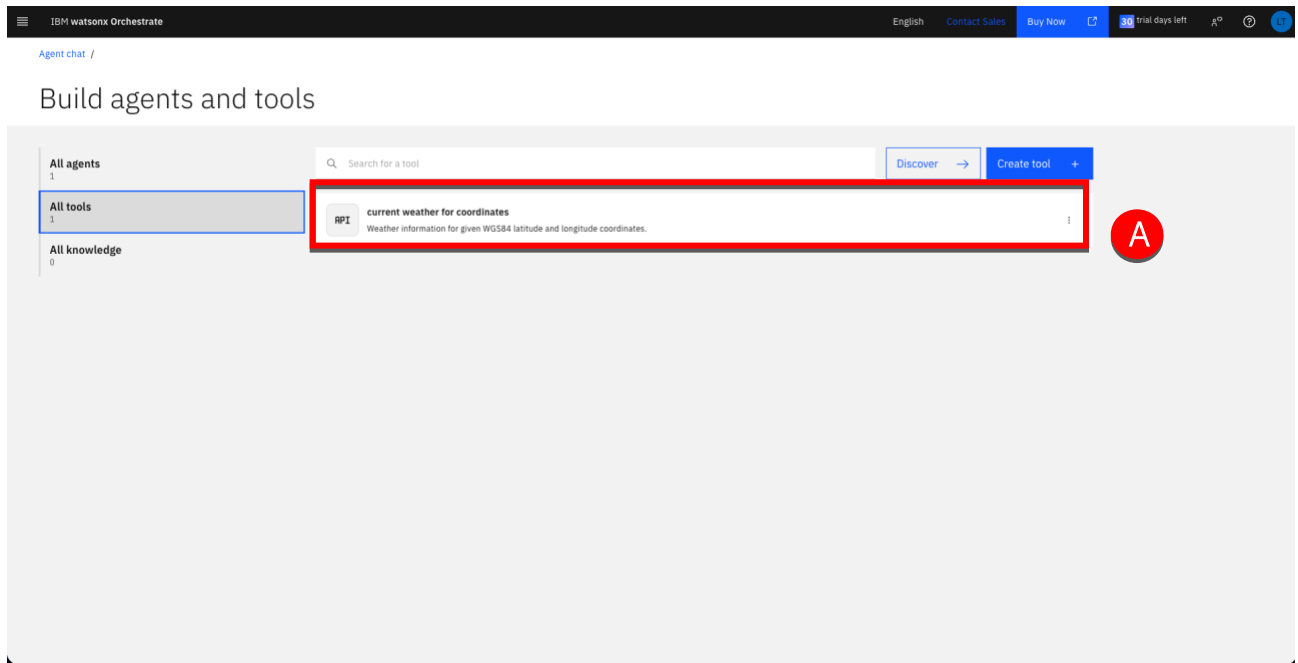
5. Click the **All tools** menu (A).



6. Type 'forecast' in the search bar if you have numerous tools in your instance (A)



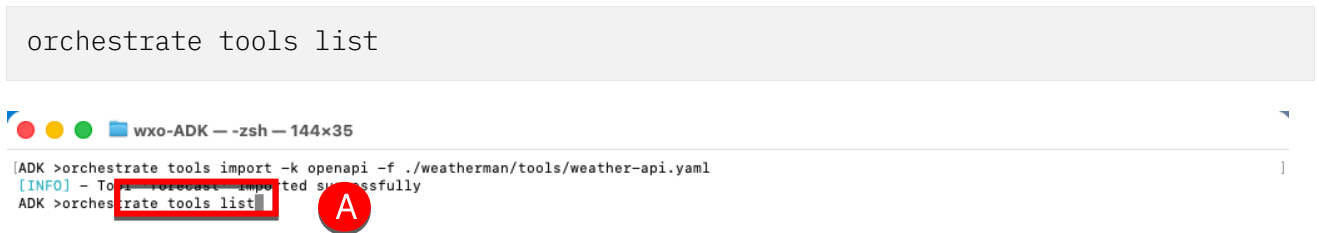
7. You can observe that the forecast tool appears in the list.



View the list of tools

The ADK also let you inspect all the tools already deployed in your SaaS instance.

1. In your ADK environment, type the following command in your console (A)



- You can see the list of all available tools in your SaaS environment (A).

```

[Please enter WKO API key:
[INFO] - Environment 'wxOSaaS' is now active
[INFO] - Active workspace: Global workspace
ADK >orchestrate tools import -k openapi -f ./weatherman/tools/weather-api.yaml
[INFO] - Tool 'forecast' imported successfully
ADK >orchestrate tools list

```

Name	Description	Type	Toolkit	App ID
Customers_5627TJ (Customers)	Retrieves a list of all customers.	openapi		
Customers_by_ID_ID_63838F (Customers by ID)	Retrieves a specific customers.	openapi		
Polices_by_ID_6788ve (Polices by ID)	Retrieves policy information for a specific policy.	openapi		
Customer_policies_7177Am (Customer policies)	Retrieves policy information for a specific customer.	openapi		
Customers_by_State_7712hq (Customers by State)	Retrieves a list of customers in a specific state.	openapi		
Total_premium_8876Q1 (Total premium)	Calculates the total value of policies for a customer.	openapi		
forecast (current weather for coordinates)	Weather information for given WGS84 latitude and longitude coordinates.	openapi		
i__get_flow_status_intrinsic_tool__ (Get flow status)	We can use the flow instance id to get the status of a flow. Only call this on explicit request by the user.	openapi		
i__get_schedule_intrinsic_tool__ (Get Schedules)	Use this tool to show the current schedules.	openapi		

A

Now that the tool is available in your SaaS environment, it can be used by an agent. The next section will guide you through the steps to import this agent.

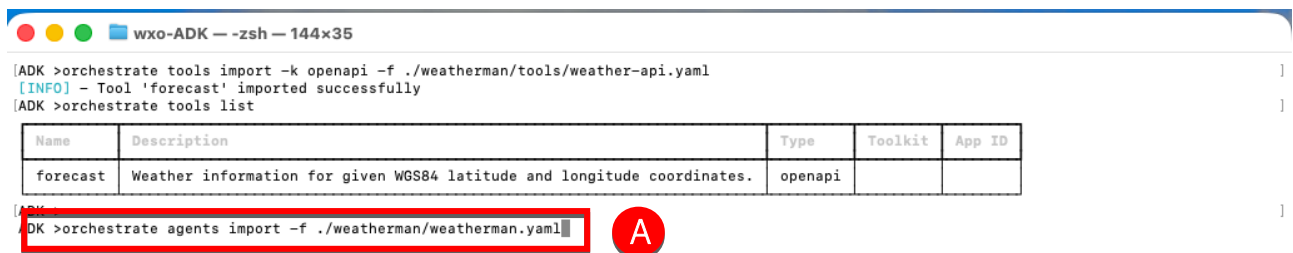
Manage agents in your SaaS tenant from your ADK

In this section, you will learn how to import agents to your SaaS environment using the watsonx Orchestrate ADK. You will recreate the two agents you may have already created in the ADK Developer Edition server, but now your commands will operate on your SaaS instance.

Import a new agent

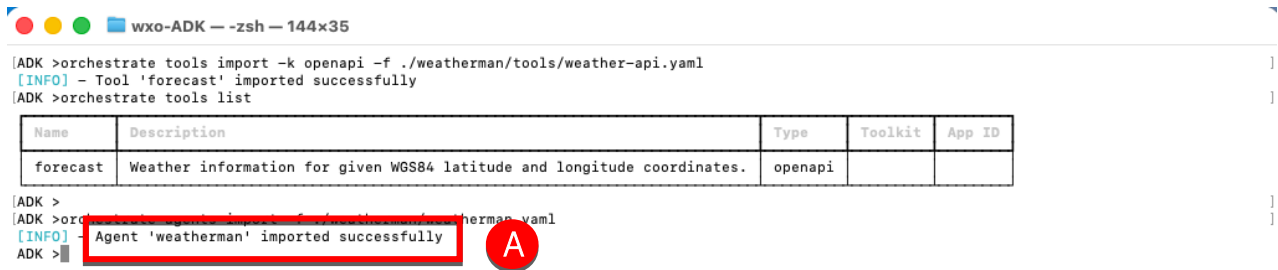
1. Type the following command in your terminal window (A).

```
orchestrate agents import -f ./weatherman/weatherman.yaml
```



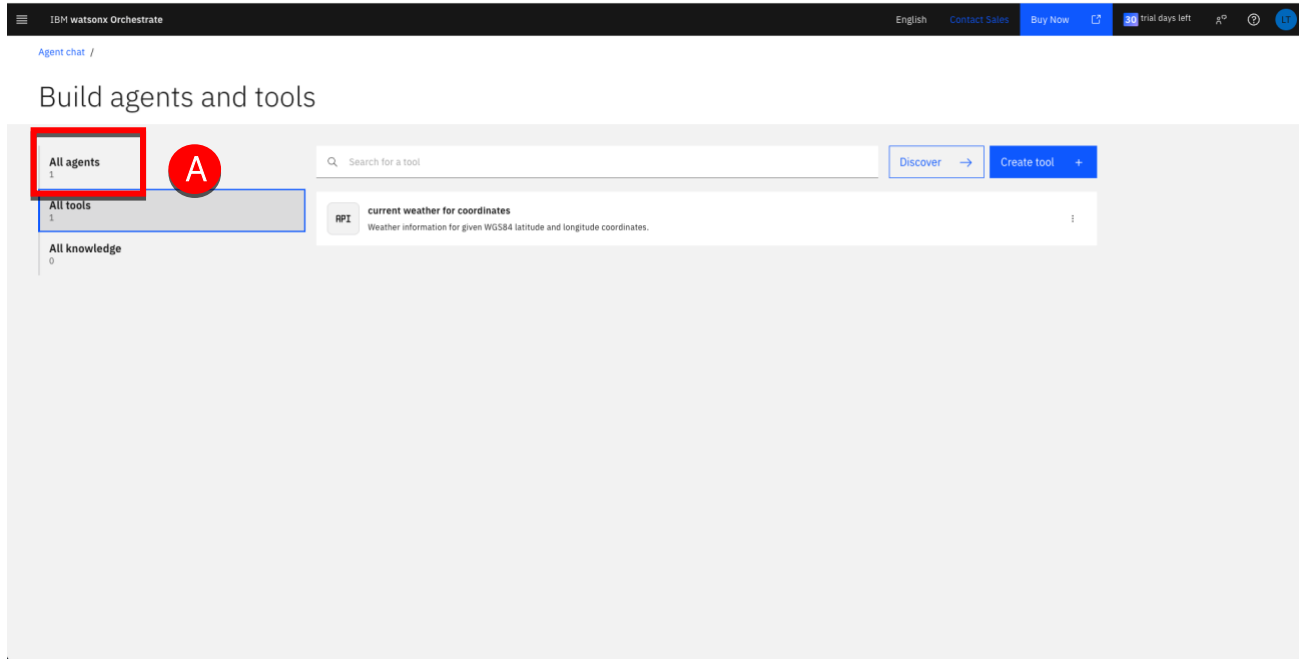
```
wxo-ADK — zsh — 144x35
[ADK >orchestrate tools import -k openapi -f ./weatherman/tools/weather-api.yaml
[INFO] - Tool 'forecast' imported successfully
[ADK >orchestrate tools list
Name      Description                                     Type      Toolkit    App ID
forecast  Weather information for given WGS84 latitude and longitude coordinates.  openapi
[ADK >orchestrate agents import -f ./weatherman/weatherman.yaml
```

2. Wait for the success confirmation message (A), then return to your SaaS environment.

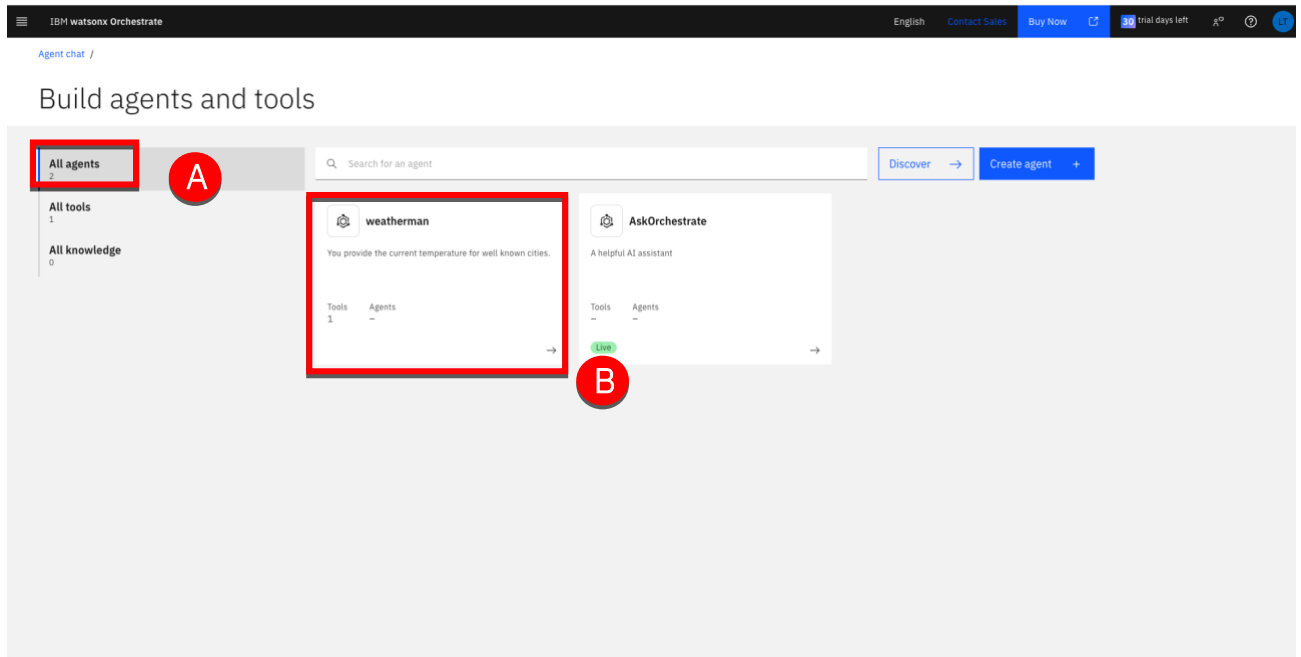


```
wxo-ADK — zsh — 144x35
[ADK >orchestrate tools import -k openapi -f ./weatherman/tools/weather-api.yaml
[INFO] - Tool 'forecast' imported successfully
[ADK >orchestrate tools list
Name      Description                                     Type      Toolkit    App ID
forecast  Weather information for given WGS84 latitude and longitude coordinates.  openapi
[ADK >orchestrate agents import -f ./weatherman/weatherman.yaml
[ADK >orchestrate agents list
[INFO] - Agent 'weatherman' imported successfully
[ADK >
```

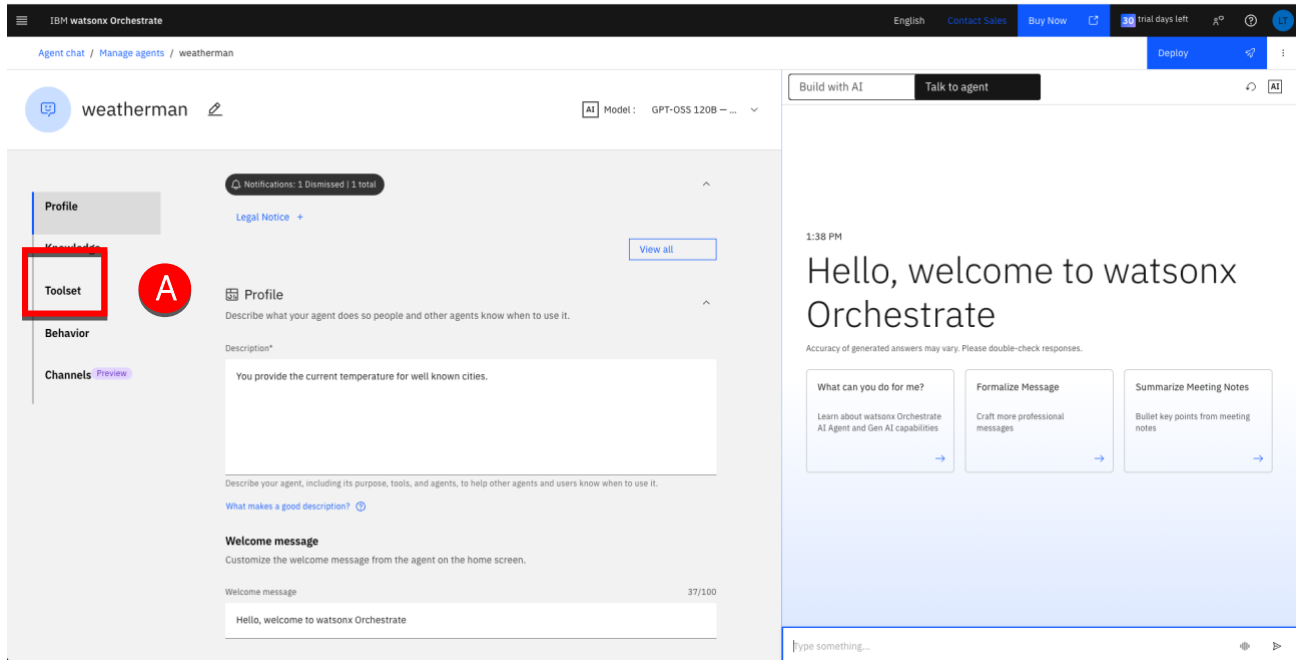
3. For now, no agent is present (assuming that you have not created any agents in the SaaS environment before starting this lab). Click Refresh your browser.



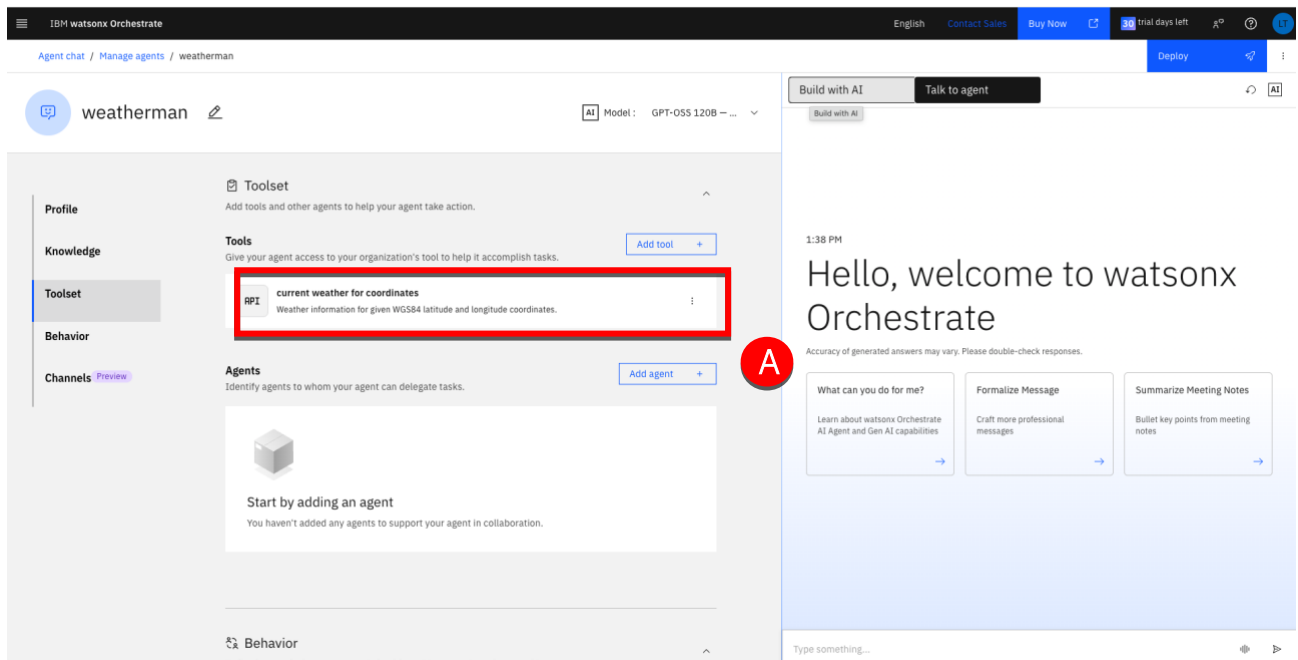
4. After a refresh, the number of agents has increased (A). The weatherman agent is now available on your SaaS environment as well. IT uses one tool. Click the **weatherman** agent tile (B).



5. Click **Toolset** (A)



6. Note that the weatherman agent uses the forecast tool (A) you have previously imported from your ADK in the previous chapter.



[View the list of agents](#)

Like the tools list, you can see all the available agent in your SaaS instance from your ADK terminal.

1. In your ADK terminal, type the following command (A)

```
orchestrate agents list
```

```
[INFO] Tool 'forecast' imported successfully  
[ADK] > orchestrate tools list
```

Name	Description	Type	Toolkit	App ID
forecast	Weather information for given WGS84 latitude and longitude coordinates.	openapi		

```
[ADK] >  
[ADK] > orchestrate agents import -f ./weatherman/weatherman.yaml  
[INFO] - Agent 'weatherman' imported successfully  
[ADK] >  
[ADK] > orchestrate agents list
```



2. Note that the **weatherman** agent is listed in the native agents table (first table).

```
wxo-ADK --zsh -- 144x35
```

Name	Description	Type	Toolkit	App ID
forecast	Weather information for given WGS84 latitude and longitude coordinates.	openapi		

```
[ADK] >  
[ADK] > orchestrate agents import -f ./weatherman/weatherman.yaml  
[INFO] - Agent 'weatherman' imported successfully  
[ADK] >  
[ADK] > orchestrate agents list
```

Agents

Name	Description	LLM	Style	Collaborators	Tools	Knowledge Base	ID
weatherman	You provide the current temperature for well known cities.	groq/openai/gpt-oss-120b	react		forecast		9c444914-ced2-4fb3-98df-0450039f1e27
AskOrchestrate	A helpful AI assistant	watsonx/meta-llama/llama-3-2-90b-vision-instruct	default				8cfc61f0-5f3b-47d9-ah48-02bc1faf75ae



External Agents

Name	Title	Description	Tags	API URL	Chat Params	Config	Nickname	App ID	ID

Assistant Agents

Name	Title	Description	Tags	Nickname	CRN	Instance URL	Assistant ID	Environment ID	ID

```
ADK >
```

Note this command also displays internal and assistant agents that may be available on your SaaS instance.

Import a knowledge base agent

In the previous step, you imported the Weatherman agent, which uses the Forecast tool. Similarly, you can import a knowledge base agent using the ADK. To do this, first import the knowledge base, then the agent - just as you may have done in a previous ADK lab.

1. Enter the following command into the terminal (A) to create the knowledge base in your SaaS environment.

```
orchestrate knowledge-bases import -f
ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml
```

```
[ADK >
[ADK >orchestrate agents import -f ./weatherman/weatherman.yaml
[INFO] - Agent 'weatherman' imported successfully
[ADK >
[ADK >orchestrate agents list
```

Name	Description	LLM	Style	Collaborators	Tools	Knowledge Base	ID
weatherman	You provide the current temperature for well known cities.	groq/openai/gpt-oss-120b	react		forecast		9c444914-ced2-4fb3-98df-0450039f1e27
AskOrchestrate	A helpful AI assistant	watsonx/meta-llama/llama-3-2-90b-vision-instruct	default				8cfc61f0-5f3b-47d9-ab48-02bc1faf75ae

Name	Title	Description	Tags	API URL	Chat Params	Config	Nickname	App ID	ID

Name	Title	Description	Tags	Nickname	CRN	Instance URL	Assistant ID	Environment ID	ID

```
[ADK >
[ADK >orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml]
```

A

Note: for more information on the knowledge based ADK, please refer to the 'ADK – Knowledge base' lab in this wxO L4 plan.

2. You can check the status of the knowledge base. Type the following command in your terminal (A).

```
orchestrate knowledge-bases status -n ibm_knowledge_base
```

```

[ADK >orchestrate env list
local http://localhost:4321 (active)
[ADK >orchestrate env add -n wxOSaaS -u https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/71cf0f6b-18c1-4b81-ac36-45e7ebbfccc4 --type ibm_iam --acti]
vate
[INFO] - Environment 'wxOSaaS' has been created
Please enter WXO API key:
[INFO] - Environment 'wxOSaaS' is now active
[ADK >orchestrate tools import -k openapi -f ./weatherman/tools/weather-api.yaml
[INFO] - Tool 'forecast' imported successfully
[ADK >orchestrate agents import -f ./weatherman/weatherman.yaml
[INFO] - Agent 'weatherman' imported successfully
[ADK >orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml
✓ Successfully imported knowledge base 'ibm_knowledge_base'
[ADK >orchestrate knowledge-bases status -n ibm_knowledge_base

```

Name	Description	Prioritize Built In Index	Ready	Built In Index Status	Built In Index Status Msg	Documents (2)	Id
ibm_knowledge_base	General information about IBM and its history	True	True	ready		IBM_wikipedia.pdf, history_of_ibm.pdf	3d2b9b35-5367-4ba...

```

ADK >

```

You can see the Documents contained in this knowledge-base that the agent will have access to.

3. You can now import the knowledge-based agent in your SaaS environment. Type the following command (A).

```
orchestrate agents import -f ibm_knowledge/agents/ibm_agent.yaml
```

```

[ADK >orchestrate env list
local http://localhost:4321 (active)
[ADK >orchestrate env add -n wxOSaaS -u https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/71cf0f6b-18c1-4b81-ac36-45e7ebbfccc4 --type ibm_iam --acti]
vate
[INFO] - Environment 'wxOSaaS' has been created
Please enter WXO API key:
[INFO] - Environment 'wxOSaaS' is now active
[ADK >orchestrate tools import -k openapi -f ./weatherman/tools/weather-api.yaml
[INFO] - Tool 'forecast' imported successfully
[ADK >orchestrate agents import -f ./weatherman/weatherman.yaml
[INFO] - Agent 'weatherman' imported successfully
[ADK >orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml
✓ Successfully imported knowledge base 'ibm_knowledge_base'
[ADK >orchestrate knowledge-bases status -n ibm_knowledge_base

```

Name	Description	Prioritize Built In Index	Ready	Built In Index Status	Built In Index Status Msg	Documents (2)	Id
ibm_knowledge_base	General information about IBM and its history	True	True	ready		IBM_wikipedia.pdf, history_of_ibm.pdf	3d2b9b35-5367-4ba...

```

[ADK >orchestrate agents import -f ibm_knowledge/agents/ibm_agent.yaml

```

4. Wait for the success message (A) and then type 'orchestrate agents list' (B) to see the imported agents into your active SaaS environment.

```
wxo-ADK --zsh -- 161x39

[ADK >orchestrate env list
local http://localhost:4321 (active)
[ADK >orchestrate env add -n wxOSaaS -u https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/71cf0f6b-18c1-4b81-ac36-45e7ebbfccc4 --type ibm_iam --active
[INFO] - Environment 'wxOSaaS' has been created
Please enter WXO API key:
[INFO] - Environment 'wxOSaaS' is now active
[ADK >orchestrate tools import -k openapi -f ./weatherman/tools/weather-api.yaml
[INFO] - Tool 'forecast' imported successfully
[ADK >orchestrate agents import -f ./weatherman/weatherman.yaml
[INFO] - Agent 'weatherman' imported successfully
[ADK >orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml
✓ Successfully imported knowledge base 'ibm_knowledge_base'
[ADK >orchestrate knowledge-bases status -n ibm_knowledge_base
```

Name	Description	Prioritize Built In Index	Ready	Built In Index Status	Built In Index Status Msg	Documents (2)	Id
ibm_knowledge_base	General information about IBM and its history	True	True	ready		IBM_wikipedia.pdf, history_of_ibm.pdf	3d2b9b35-5367-4ba...

```
[ADK >orchestrate agents import -f ibm_knowledge/agents/ibm_agent.yaml
[INFO] - Agent 'ibm_agent' imported successfully
[ADK >orchestrate agents list
```

(A) (B)

5. Note that the ibm_agent is now available in the native agents in your SaaS environment. It uses the knowledge base previously imported. You can also see it directly from your SaaS environment UI.

```
wxo-ADK --zsh -- 144x35

[ADK >orchestrate agents import -f ibm_knowledge/agents/ibm_agent.yaml
[INFO] - Agent 'ibm_agent' imported successfully
[ADK >
[ADK >orchestrate agents list
```

Name	Description	LLM	Style	Collaborators	Tools	Knowledge Base	ID
ibm_agent	You are an helpful agent that has knowledge about IBM and its history	groq/openai/gpt-oss-120b	react			ibm_knowledge_base	70c51459-aad6-4df8-a234-4c3481810186
weatherman	You provide the current temperature for well known cities.	groq/openai/gpt-oss-120b	react		forecast		9c444914-ced2-4fb3-98df-0450839f1e27
AskOrchestrate	A helpful AI assistant	watsonx/meta-llama/1lama-3-2-90b-vision-instruct	default				8cfc61f0-5f3b-47d9-ab48-02bc1faf75ae

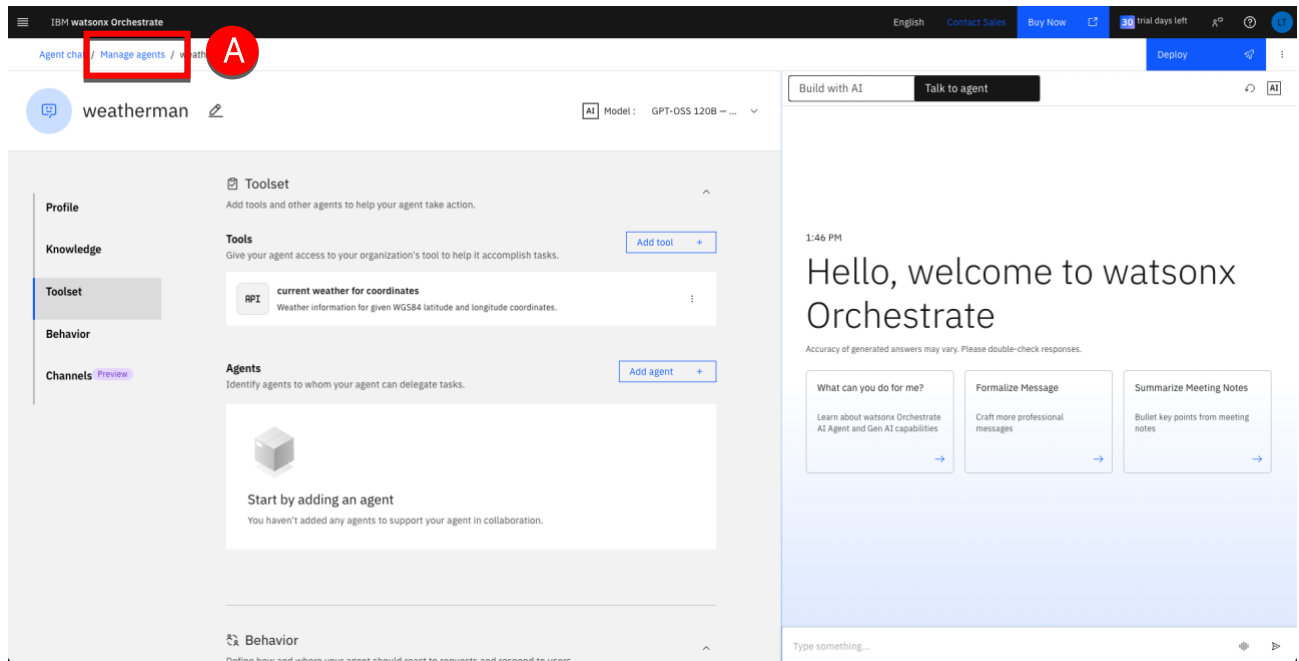
(A)

Name	Title	Description	Tags	API URL	Chat Params	Config	Nickname	App ID	ID
------	-------	-------------	------	---------	-------------	--------	----------	--------	----

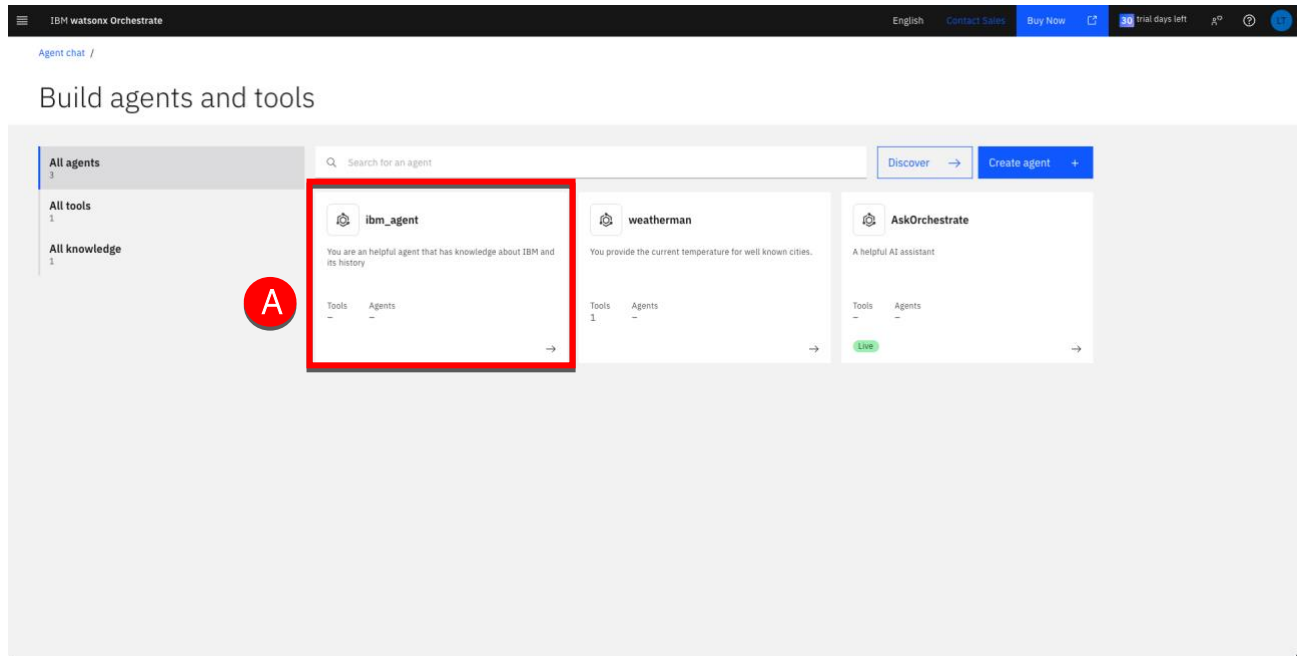
Name	Title	Description	Tags	Nickname	CRN	Instance URL	Assistant ID	Environment ID	ID
------	-------	-------------	------	----------	-----	--------------	--------------	----------------	----

```
ADK >
```

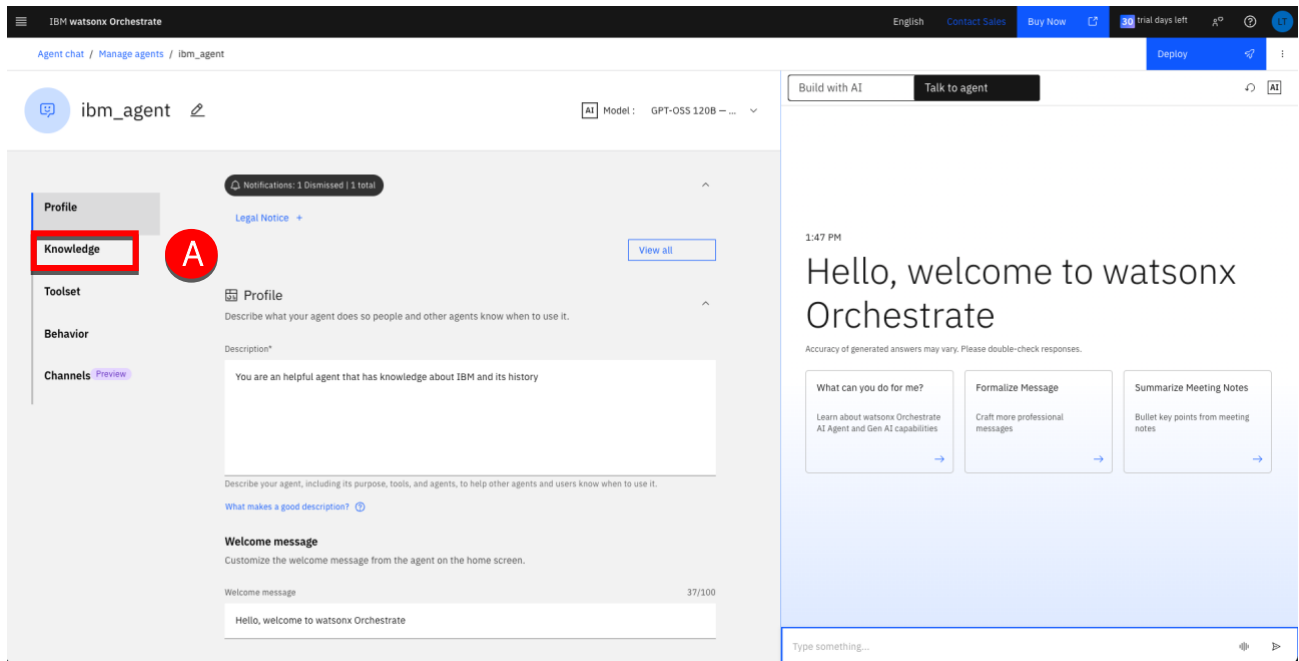
6. Return to your SaaS environment and click on the **Manage agents** navigation menu (A).



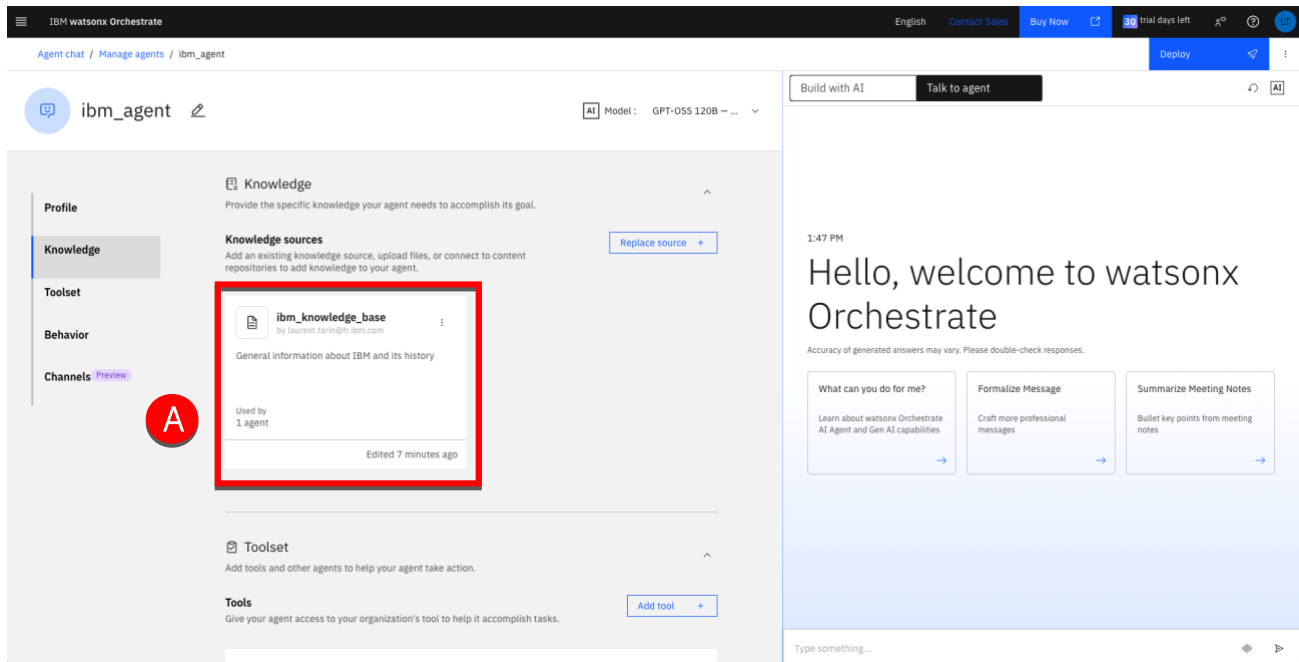
7. The `ibm_agent` is there, click the `ibm_agent` tile (A).



8. Click Knowledge (A).



9. Note the agent is using the imported knowledge-base (A) from the ADK.



10. Click the **ibm_knowledge_base** menu icon (A) then click **Edit details** (B).

IBM watsonx Orchestrate

Agent chat / Manage agents / ibm_agent

ibm_agent

Model: GPT-OS5 120B

Knowledge

Provide the specific knowledge your agent needs to accomplish its goal.

Knowledge sources

Add an existing knowledge source, upload files, or connect to external repositories to add knowledge to your agent.

ibm_knowledge_base

Generated by IBM watsonx Orchestrate

Edit details

Remove

Used by 1 agent

Edited 6 minutes ago

Toolset

Add tools and other agents to help your agent take action.

Tools

Give your agent access to your organization's tool to help it accomplish tasks.

Replace source +

Add tool +

Build with AI

Talk to agent

Deploy

1:49 PM

Hello, welcome to watsonx Orchestrate

Accuracy of generated answers may vary. Please double-check responses.

What can you do for me?
Learn about watsonx Orchestrate AI Agent and Gen AI capabilities

Formalize Message
Craft more professional messages

Summarize Meeting Notes
Bullet key points from meeting notes

Type something...

11. Note the 2 files contained in the knowledge base (A). Click **ibm_agent** (B).

IBM watsonx Orchestrate

Agent chat / Manage agents / **ibm_agent** / ibm_knowledge_base

ibm_knowledge_base

Delete

Files

Description

General information about IBM and its history

Detailed descriptions will help guide your agents to effectively use your knowledge.

Edit knowledge settings

Documents (2)

File name	URL source	Date uploaded
IBM_wikipedia.pdf		1/28/26
history_of_ibm.pdf		1/28/26

Add documents +

Learn more about best knowledge practices

Manage Connections in your SaaS tenant from your ADK

The ADK also let you manage the connections. At the time we are writing this lab, the only way to remove connections from a SaaS tenant is through the ADK.

This chapter will not go into the details of how to manage the connections configuration and how to set up the draft and live environments since this aspect is covered in the Architecture – Build an agent with MCP tools lab part 2.

You can manage all the existing connections in your SaaS instance using the connections ADK command.

View the list of existing connections

1. From your ADK terminal, type 'orchestrate connections list' (A).

wxo-ADK — zsh — 144x35

[INFO] - Agent 'ibm_agent' imported successfully

ADK >

ADK >orchestrate agents list

Agents

Name	Description	LLM	Style	Collaborators	Tools	Knowledge Base	ID
ibm_agent	You are an helpful agent that has knowledge about IBM and its history	groq/openai/gpt-oss-120b	react			ibm_knowledge_base	70c51459-aad6-4df8-a234-4c3481810186
weatherman	You provide the current temperature for well known cities.	groq/openai/gpt-oss-120b	react		forecast		9c444914-ced2-4fb3-98df-0450039f1e27
AskOrchestrate	A helpful AI assistant	watsonx/meta-llama/1lama-3-2-90b-vision-instruct	default				8cfc61f0-5f3b-47d9-ab48-02bc1faf75ae

External Agents

Name	Title	Description	Tags	API URL	Chat Params	Config	Nickname	App ID	ID

Assistant Agents

Name	Title	Description	Tags	Nickname	CRN	Instance URL	Assistant ID	Environment ID	ID

ADK >

ADK >orchestrate connections list

A

2. Scroll-up (A) and note the different Live and draft connections defined in your SaaS environment.

wxo-ADK — -zsh — 144x35

oracle_hcm_key_value_ibm_184bdbd3	n/a	n/a	×
sap_successfactors_basic_ibm_184bdbd3	n/a	n/a	×
microsoft_key_value_ibm_184bdbd3	n/a	n/a	×
oracle_hcm_basic_ibm_184bdbd3	n/a	n/a	×
servicenow_key_value_ibm_184bdbd3	n/a	n/a	×
google_key_value_ibm_184bdbd3	n/a	n/a	×
servicenow_bearer_ibm_184bdbd3	n/a	n/a	×
sap_successfactors_key_value_ibm_184bdbd3	n/a	n/a	×
jira_key_value_ibm_184bdbd3	n/a	n/a	×
slack_key_value_ibm_184bdbd3	n/a	n/a	×
dnb_procurement_key_value_ibm_184bdbd3	n/a	n/a	×
dnb_sales_key_value_ibm_184bdbd3	n/a	n/a	×
dnb_base_key_value_ibm_184bdbd3	n/a	n/a	×
box_key_value_ibm_184bdbd3	n/a	n/a	×
jira_basic_ibm_184bdbd3	n/a	n/a	×
coupa_key_value_ibm_184bdbd3	n/a	n/a	×
adobe_workfront_key_value_ibm_184bdbd3	n/a	n/a	×

ADK >



3. Note that all connections are grouped into 2 tables. One for the Draft environment (A), and another one below for the Live one.

wxo-ADK — -zsh — 144x35

External Agents									
Name	Title	Description	Tags	API URL	Chat Params	Config	Nickname	App ID	ID

Assistant Agents									
Name	Title	Description	Tags	Nickname	CRN	Instance URL	Assistant ID	Environment ID	ID

ADK >

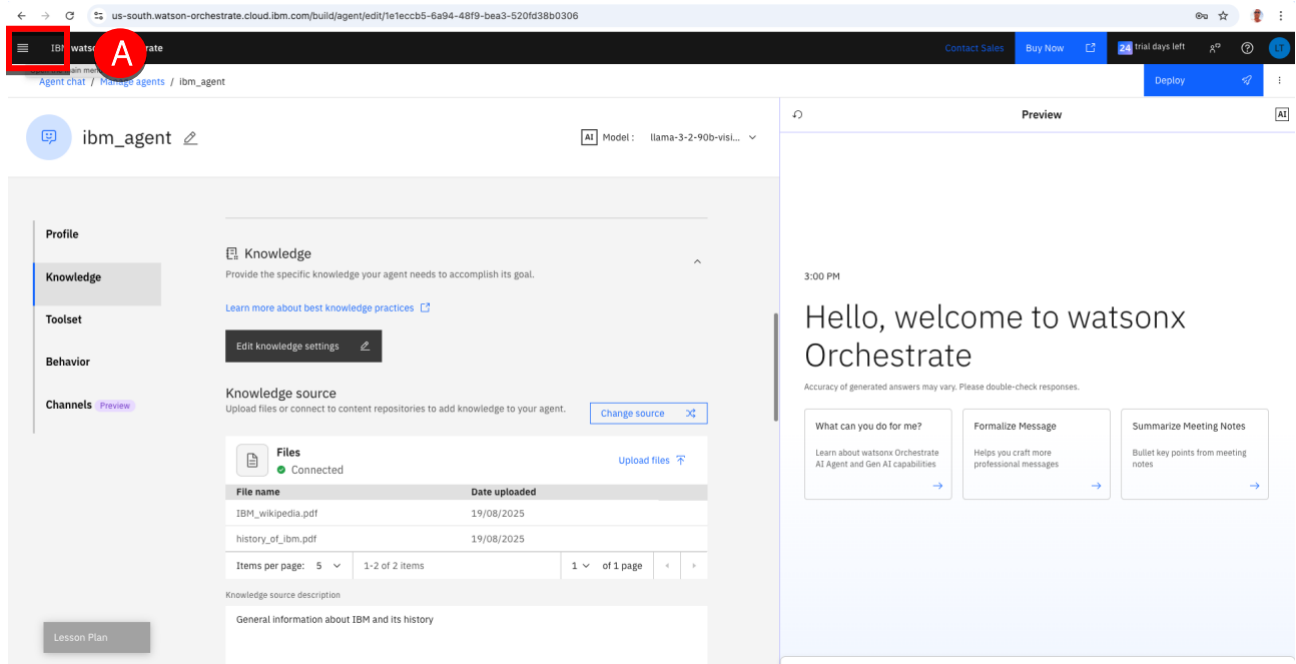
ADK >orchestrate connections list

A **Draft**

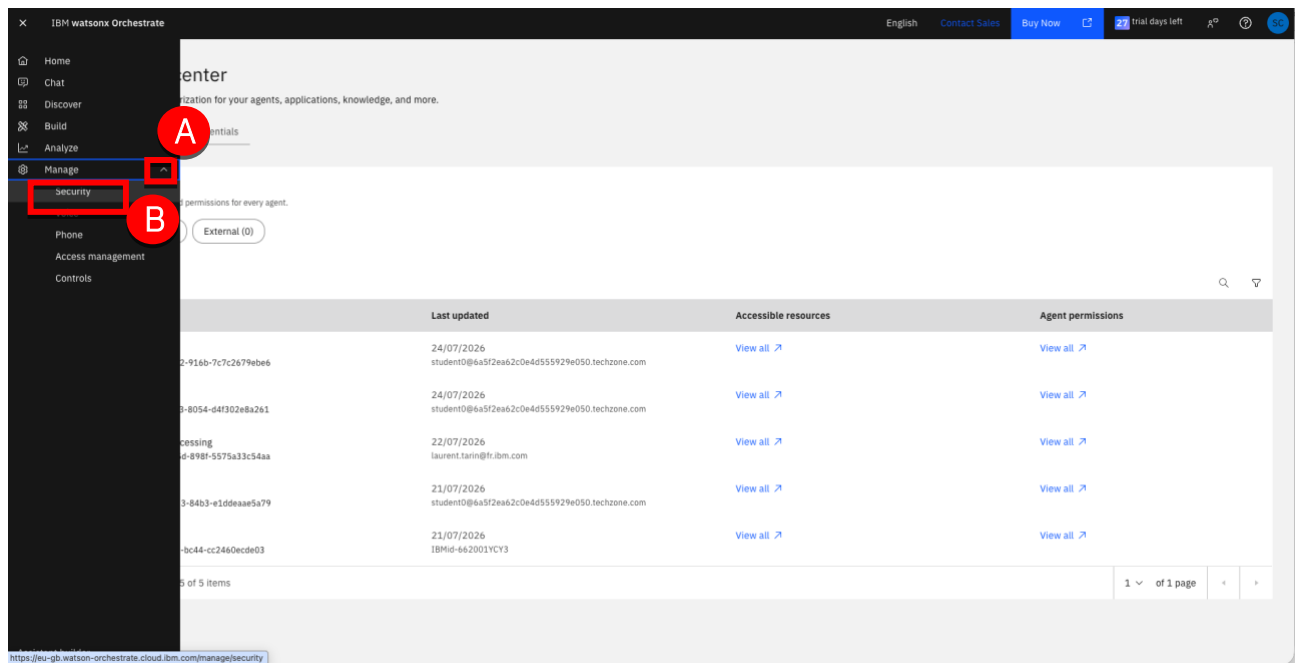
App ID	Auth Type	Type	Credentials Set/ Connected
nativemsg_rcs_nativemsg_15626a79	n/a	n/a	×
biw_nomination_biw_c539a2d8	n/a	n/a	×
testrail_ibm_184bdbd3	n/a	n/a	×
zendesk_ibm_184bdbd3	n/a	n/a	×
maximo_it_ibm_184bdbd3	n/a	n/a	×
red_hat_ansible_ibm_184bdbd3	n/a	n/a	×
ariba_master_data_retrieval_ibm_184bdbd3	n/a	n/a	×
dynamics_365_ibm_184bdbd3	n/a	n/a	×
uptok_api_key_uptok_1917c351	n/a	n/a	×

Note: Connections credentials and connection status is also displayed for each connection.

4. Return to your SaaS environment and click the hamburger icon (A);



5. Expand the **Manage** menu (A), select **Security** (B).



6. Select the **Connections** tab (A).

The screenshot shows the IBM watsonx Orchestrate Security control center interface. The top navigation bar includes 'English', 'Contact Sales', 'Buy Now', and 'Trial days left'. The main header is 'Security control center' with the subtitle 'Manage authentication and authorization for your agents, applications, knowledge, and more.' Below this, there are three tabs: 'Agents', 'Connections' (which is selected and highlighted with a red box and a red circle labeled 'A'), and 'Team credentials'. The 'Agent access overview' section shows filters for 'Native (5)', 'Imported (0)', and 'External (0)'. A table lists agents with columns for 'Agent', 'Last updated', 'Accessible resources', and 'Agent permissions'. The table contains five rows of agent information. At the bottom, there are pagination controls showing 'Items per page: 10' and '1-5 of 5 items'.

Agent	Last updated	Accessible resources	Agent permissions
ibm_agent 64503427-822b-4f12-916b-7c7c2679eb6e	24/07/2026 student0@6a5f2ea82c0e4d555929e050.techzone.com	View all	View all
weatherman 2cbe9a83-0a7a-4223-8054-d4f302e8a261	24/07/2026 student0@6a5f2ea82c0e4d555929e050.techzone.com	View all	View all
LOT Document processing 44081799-7a9e-475d-898f-5575a33c54aa	22/07/2026 laurent.tarin@fr.ibm.com	View all	View all
Insurance e1d5a55d-1465-4123-84b3-e1ddea9e5a79	21/07/2026 student0@6a5f2ea82c0e4d555929e050.techzone.com	View all	View all
AskOrchestrate cf9d1c0e-3830-4f0d-bc44-cc2460ecde03	21/07/2026 IBMId-6a2001YCY3	View all	View all

7. Click **Pre-built** (A).

The screenshot shows the IBM watsonx Orchestrate Security control center interface, specifically the 'Connections' tab. The top navigation bar is the same as in the previous screenshot. The 'Connections' tab is selected and highlighted with a red box and a red circle labeled 'A'. Below the tab, there is a section titled 'Connections' with the subtitle 'Add connections to integrate with third-party applications. Learn more'. There are two buttons: 'Custom (0)' and 'Pre-built (152)', with 'Pre-built' being selected and highlighted with a red box. Below this, there is a table with columns for 'Name', 'Draft', 'Live', and 'Last updated'. The table is currently empty, showing 'No connections' and the message 'You have not added connections yet'. There is an 'Add connection' button with a plus sign.

Name	Draft	Live	Last updated
------	-------	------	--------------

- Review all the existing connections (A). The corresponds to the one you have listed from your ADK terminal using the connections list command.

Security control center

Manage authentication and authorization for your agents, applications, knowledge, and more.

Agents **Connections** Team credentials

Connections

Add connections to integrate with third-party applications. [Learn more](#)

Custom (0) Pre-built (152)

Name	Draft	Live	Last updated	
Viclinic Inc viclinic_key_value_viclinic_de530...	Configuration pending	Configuration pending	7/17/2026	!
Gemini gemini_web_search_ibm_184bdb...	Configuration pending	Configuration pending	7/17/2026	!
IBM Watson Data Intelligence... data_intelligence_mcp_server_ib...	Configuration pending	Configuration pending	6/29/2026	!
IBM Carbon MCP carbon_mcp_ibm_184bdb3	Configuration pending	Configuration pending	6/29/2026	!
IBM Confluent Cloud Globa... confluent_cloud_global_ibm_184...	Configuration pending	Configuration pending	6/29/2026	!
IBM Confluent Cloud Regio... confluent_cloud_regional_ibm_1...	Configuration pending	Configuration pending	6/29/2026	!

Create a new connection

In this section you will create connection using the ADK. You will learn in the Architecture – Build an agent using MCP tools lab part 2 how to precisely configure the different environments using the ADK

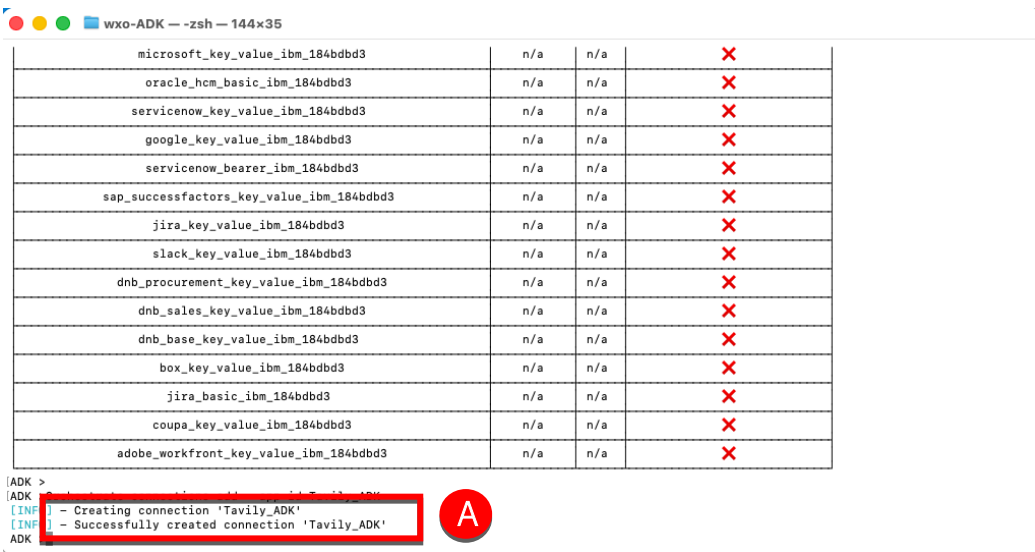
- In your ADK terminal, type the following command to create a connection with the application id name Tavily_ADK. The ADK suffix is not significant, it just distinguishes this connection from any other Tavily connection created in an earlier lab.

```
orchestrate connections add --app-id Tavily_ADK
```

viclinic_key_value_ibm_184bdb3	n/a	n/a	×
servicenow_key_value_ibm_184bdb3	n/a	n/a	×
google_key_value_ibm_184bdb3	n/a	n/a	×
servicenow_bearer_ibm_184bdb3	n/a	n/a	×
sap_successfactors_key_value_ibm_184bdb3	n/a	n/a	×
jira_key_value_ibm_184bdb3	n/a	n/a	×
slack_key_value_ibm_184bdb3	n/a	n/a	×
dnb_procurement_key_value_ibm_184bdb3	n/a	n/a	×
dnb_sales_key_value_ibm_184bdb3	n/a	n/a	×
dnb_base_key_value_ibm_184bdb3	n/a	n/a	×
box_key_value_ibm_184bdb3	n/a	n/a	×
jira_basic_ibm_184bdb3	n/a	n/a	×
coupa_key_value_ibm_184bdb3	n/a	n/a	×
adobe_workfront_key_value_ibm_184bdb3	n/a	n/a	×

```
ADK >
ADK >Orchestrate connections add --app-id Tavily_ADK
```

- Wait for the success notification message (A).



The terminal window shows a table of failed API calls and a successful connection creation message. The table has columns for the API call name, the response, and a status indicator. The status indicator is a red 'X' for failed calls and a green checkmark for successful calls. The successful call is 'Tavily_ADK'.

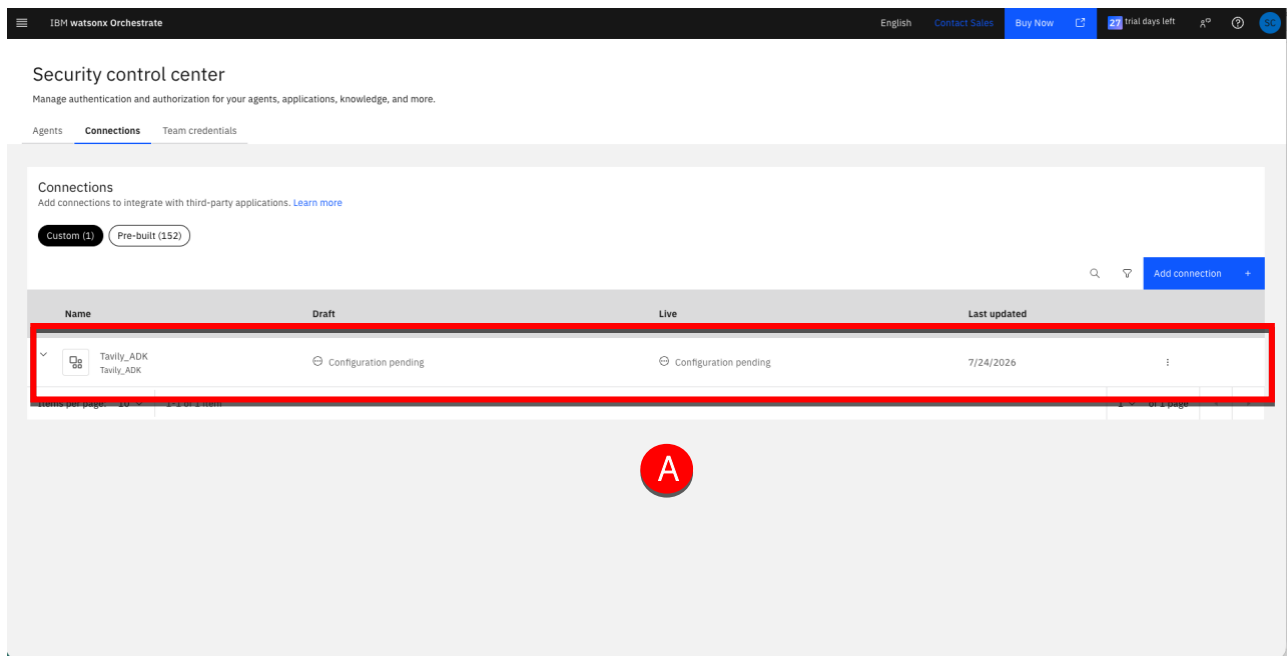
API Call	Response	Status
microsoft_key_value_ibm_184bdbd3	n/a	✗
oracle_hcm_basic_ibm_184bdbd3	n/a	✗
servicenow_key_value_ibm_184bdbd3	n/a	✗
google_key_value_ibm_184bdbd3	n/a	✗
servicenow_bearer_ibm_184bdbd3	n/a	✗
sap_successfactors_key_value_ibm_184bdbd3	n/a	✗
jira_key_value_ibm_184bdbd3	n/a	✗
slack_key_value_ibm_184bdbd3	n/a	✗
dnb_procurement_key_value_ibm_184bdbd3	n/a	✗
dnb_sales_key_value_ibm_184bdbd3	n/a	✗
dnb_base_key_value_ibm_184bdbd3	n/a	✗
box_key_value_ibm_184bdbd3	n/a	✗
jira_basic_ibm_184bdbd3	n/a	✗
coupa_key_value_ibm_184bdbd3	n/a	✗
adobe_workfront_key_value_ibm_184bdbd3	n/a	✗

Below the table, the terminal shows the following messages:

```
ADK >
ADK >
[INF] - Creating connection 'Tavily_ADK'
[INF] - Successfully created connection 'Tavily_ADK'
ADK >
```

A red circle with the letter 'A' is placed next to the successful connection creation message.

- Return to your SaaS environment. Note that the Tavily_ADK connection is there under the **Custom** tab (A). Refresh your browser. No connections details have been provided for the Draft and live environments.



The screenshot shows the IBM Watsonx Orchestrate Security control center. The 'Connections' tab is selected, and the 'Custom (1)' filter is applied. The table shows one connection, 'Tavily_ADK', which is in the 'Draft' state. The 'Live' state is also shown as 'Configuration pending'. The 'Last updated' date is 7/24/2026. A red circle with the letter 'A' is placed below the table.

Name	Draft	Live	Last updated
Tavily_ADK	Configuration pending	Configuration pending	7/24/2026

If your Tavily_ADK environment is not shown, try logging into IBM Cloud with a private browser and opening your Orchestrate instance.

Remove elements from your SaaS tenant using the ADK

In this chapter, you will learn how to remove elements from a SaaS tenant using ADK commands. Let's clean our SaaS tenant from the elements created during this lab.

Remove a connection

The only way to remove a connection from a SaaS tenant (at the time of writing of this manual) is to use the ADK.

1. In your ADK terminal, type the following command (A).

```
orchestrate connections remove -a Tavily_ADK
```

jira_key_value_ibm_184bdbd3	n/a	n/a	✗
slack_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_procurement_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_sales_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_base_key_value_ibm_184bdbd3	n/a	n/a	✗
box_key_value_ibm_184bdbd3	n/a	n/a	✗
jira_basic_ibm_184bdbd3	n/a	n/a	✗
coupa_key_value_ibm_184bdbd3	n/a	n/a	✗
adobe_workfront_key_value_ibm_184bdbd3	n/a	n/a	✗

```
[ADK >
[ADK >Orchestrate connections add --app-id Tavily_ADK
[INFO] - Creating connection 'Tavily_ADK'
[INFO] - Successfully created connection 'Tavily_ADK'
[ADK >
[ADK >orchestrate connections remove -a Tavily_ADK
```



2. Wait for the confirmation message (A).

slack_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_procurement_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_sales_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_base_key_value_ibm_184bdbd3	n/a	n/a	✗
box_key_value_ibm_184bdbd3	n/a	n/a	✗
jira_basic_ibm_184bdbd3	n/a	n/a	✗
coupa_key_value_ibm_184bdbd3	n/a	n/a	✗
adobe_workfront_key_value_ibm_184bdbd3	n/a	n/a	✗

```
[ADK >
[ADK >Orchestrate connections add --app-id Tavily_ADK
[INFO] - Creating connection 'Tavily_ADK'
[INFO] - Successfully created connection 'Tavily_ADK'
[ADK >
[ADK >orchestrate connections remove -a Tavily_ADK
[INFO] - Removing connection 'Tavily_ADK'
[INFO] - Connection 'Tavily_ADK' successfully removed
[ADK >
```



Note: You can return to your SaaS environment to check the deletion of your Tavili connection.

Remove a knowledge base

The only way to remove a knowledge-base from a SaaS environment at the time we are writing this guide is to use the ADK.

1. To remove a knowledge base for the environment, enter the following command (A).

```
orchestrate knowledge-bases remove -n ibm_knowledge_base
```

jira_key_value_ibm_184bdbd3	n/a	n/a	✗
slack_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_procurement_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_sales_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_base_key_value_ibm_184bdbd3	n/a	n/a	✗
box_key_value_ibm_184bdbd3	n/a	n/a	✗
jira_basic_ibm_184bdbd3	n/a	n/a	✗
coupa_key_value_ibm_184bdbd3	n/a	n/a	✗
adobe_workfront_key_value_ibm_184bdbd3	n/a	n/a	✗

```
[ADK >
[ADK >Orchestrate connections add --app-id Tavily_ADK
[INFO] - Creating connection 'Tavily_ADK'
[INFO] - Successfully created connection 'Tavily_ADK'
[ADK >
[ADK >orchestrate connections remove -a Tavily_ADK
[INFO] - Removing connection 'Tavily_ADK'
[INFO] - Connection 'Tavily_ADK' successfully removed
[ADK >orchestrate knowledge-bases remove -n ibm_knowledge_base
```

A

2. Wait for the removal confirmation message (A).

wxo-ADK — zsh — 144x35

google_key_value_ibm_184bdbd3	n/a	n/a	✗
servicenow_bearer_ibm_184bdbd3	n/a	n/a	✗
sap_successfactors_key_value_ibm_184bdbd3	n/a	n/a	✗
jira_key_value_ibm_184bdbd3	n/a	n/a	✗
slack_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_procurement_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_sales_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_base_key_value_ibm_184bdbd3	n/a	n/a	✗
box_key_value_ibm_184bdbd3	n/a	n/a	✗
jira_basic_ibm_184bdbd3	n/a	n/a	✗
coupa_key_value_ibm_184bdbd3	n/a	n/a	✗
adobe_workfront_key_value_ibm_184bdbd3	n/a	n/a	✗

```
[ADK >
[ADK >Orchestrate connections add --app-id Tavily_ADK
[INFO] - Creating connection 'Tavily_ADK'
[INFO] - Successfully created connection 'Tavily_ADK'
[ADK >
[ADK >orchestrate connections remove -a Tavily_ADK
[INFO] - Removing connection 'Tavily_ADK'
[INFO] - Connection 'Tavily_ADK' successfully removed
[ADK >orchestrate knowledge-bases remove -n ibm_knowledge_base
[INFO] - Successfully removed knowledge base 'ibm_knowledge_base'
[ADK >
```

A

Remove an agent

To remove an agent with the ADK, you must specify the type of agent (Native / Internal / Assistant) and the name of the agent to remove.

1. In your ADK terminal, type the following command (A).

```
orchestrate agents remove -k native -n weatherman
```

dnb_base_key_value_ibm_184bdbd3	n/a	n/a	✗
box_key_value_ibm_184bdbd3	n/a	n/a	✗
jira_basic_ibm_184bdbd3	n/a	n/a	✗
coupa_key_value_ibm_184bdbd3	n/a	n/a	✗
adobe_workfront_key_value_ibm_184bdbd3	n/a	n/a	✗

```
[ADK >
[ADK >Orchestrate connections add --app-id Tavily_ADK
[INFO] - Creating connection 'Tavily_ADK'
[INFO] - Successfully created connection 'Tavily_ADK'
[ADK >
[ADK >orchestrate connections remove -a Tavily_ADK
[INFO] - Removing connection 'Tavily_ADK'
[INFO] - Connection 'Tavily_ADK' successfully removed
[ADK >orchestrate knowledge-bases remove -n ibm_knowledge_base
[INFO] - Successfully removed knowledge base 'ibm_knowledge_base'
[ADK >
[ADK >orchestrate agents remove -k native -n weatherman
```



2. Wait for the confirmation message (A).

wxo-ADK — zsh — 144x35

sap_successfactors_key_value_ibm_184bdbd3	n/a	n/a	✗
jira_key_value_ibm_184bdbd3	n/a	n/a	✗
slack_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_procurement_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_sales_key_value_ibm_184bdbd3	n/a	n/a	✗
dnb_base_key_value_ibm_184bdbd3	n/a	n/a	✗
box_key_value_ibm_184bdbd3	n/a	n/a	✗
jira_basic_ibm_184bdbd3	n/a	n/a	✗
coupa_key_value_ibm_184bdbd3	n/a	n/a	✗
adobe_workfront_key_value_ibm_184bdbd3	n/a	n/a	✗

```
[ADK >
[ADK >Orchestrate connections add --app-id Tavily_ADK
[INFO] - Creating connection 'Tavily_ADK'
[INFO] - Successfully created connection 'Tavily_ADK'
[ADK >
[ADK >orchestrate connections remove -a Tavily_ADK
[INFO] - Removing connection 'Tavily_ADK'
[INFO] - Connection 'Tavily_ADK' successfully removed
[ADK >orchestrate knowledge-bases remove -n ibm_knowledge_base
[INFO] - Successfully removed knowledge base 'ibm_knowledge_base'
[ADK >
[ADK >orchestrate agents remove -k native -n weatherman
[INFO] - Successfully removed agent weatherman
[ADK >
```



- Type 'orchestrate agents list' (A) to list the remaining agents.

```
wxo-ADK --zsh-- 144x35
[ADK >orchestrate connections remove -a Tavily_ADK
[INFO] - Removing connection 'Tavily_ADK'
[INFO] - Connection 'Tavily_ADK' successfully removed
[ADK >orchestrate knowledge-bases remove -n ibm_knowledge_base
[INFO] - Successfully removed knowledge base 'ibm_knowledge_base'
[ADK >
[ADK >orchestrate agents remove -k native -n weatherman
[INFO] - Successfully removed agent 'weatherman'
[ADK >orchestrate agents list
```

Agents							
Name	Description	LLM	Style	Collaborators	Tools	Knowledge Base	ID
ibm_agent	You are an helpful agent that has knowledge about IBM and its history	groq/openai/gpt-oss-120b	react				70c51459-aad6-4df8-a234-4c3481810186
AskOrchestrate	A helpful AI assistant	watsonx/meta-llama/llama-3-2-90b-vision-instruct	default				8cfc61f0-5f3b-47d9-ab48-02bc1faf75ae

External Agents									
Name	Title	Description	Tags	API URL	Chat Params	Config	Nickname	App ID	ID

Assistant Agents									
Name	Title	Description	Tags	Nickname	CRN	Instance URL	Assistant ID	Environment ID	ID

```
ADK >
```

Remove a tool

In the same way, ADK commands can be used to remove tools.

- Type 'orchestrate tools list' to get the list of the tools from your SaaS environment.

Name	Description	LLM	Style	Collaborators	Tools	Knowledge Base	ID
ibm_agent	You are an helpful agent that has knowledge about IBM and its history	groq/openai/gpt-oss-120b	react				70c51459-aad6-4df8-a234-4c3481810186
AskOrchestrate	A helpful AI assistant	watsonx/meta-llama/llama-3-2-90b-vision-instruct	default				8cfc61f0-5f3b-47d9-ab48-02bc1faf75ae

External Agents									
Name	Title	Description	Tags	API URL	Chat Params	Config	Nickname	App ID	ID

Assistant Agents									
Name	Title	Description	Tags	Nickname	CRN	Instance URL	Assistant ID	Environment ID	ID

```
ADK
ADK orchestrate tools list
```

- Type the following command (A) to remove the tool named forecast.

```
orchestrate tools remove -n forecast
```

	and its history						
AskOrchestrate	A helpful AI assistant	watsonx/meta-llama/llama-3-2-90b-vision-instruct	default				8cfc61f0-5f3b-47d9-ab48-02bc1faf75ae

External Agents

Name	Title	Description	Tags	API URL	Chat Params	Config	Nickname	App ID	ID

Assistant Agents

Name	Title	Description	Tags	Nickname	CRN	Instance URL	Assistant ID	Environment ID	ID

```
[ADK >
[ADK >orchestrate tools list
```

Name	Description	Type	Toolkit	App ID
forecast	Weather information for given WGS84 latitude and longitude coordinates.	openapi		

```
[ADK >orchestrate tools remove -n forecast
```

A

- Wait for the confirmation message (A).

Name	Title	Description	Tags	Nickname	CRN	Instance URL	Assistant ID	Environment ID	ID

```
[ADK >
[ADK >orchestrate tools list
```

Name	Description	Type	Toolkit	App ID
forecast	Weather information for given WGS84 latitude and longitude coordinates.	openapi		

```
[ADK >orchestrate tools remove -n forecast
[INFO] - Successfully removed tool forecast
[ADK >
```

A

- You can return to your SaaS instance and check that the tools (A) and weatherman agent (B) have been removed.

The screenshot shows the IBM watsonx Orchestrate web interface. The top navigation bar includes 'English', 'Contact Sales', 'Buy Now', and a 'trial days left' indicator. The main heading is 'Build agents and tools'. On the left sidebar, there are two sections: 'All agents' with a count of 2, and 'All tools' with a count of 0. The 'All tools' section is highlighted with a red box and labeled 'A'. The main content area displays two agent cards. The first card is for 'ibm_agent' with the description 'You are an helpful agent that has knowledge about IBM and its history'. The second card is for 'AskOrchestrate' with the description 'A helpful AI assistant'. Both cards have 'Tools' and 'Agents' tabs. The 'AskOrchestrate' card is highlighted with a red box and labeled 'B'.

Using the '--help' in the ADK: how to remove a toolkit

During this lab, we have used different commands. Orchestrate ADK comes with a command help for each command and command options you may be interested to use. To display the help on a specific command or parameter, just add '--help' to the command.

During this lab we have not used the toolkit command since we have not yet created any toolkit unless you have already performed the Architecture – Build an agent with MCP tool lab.

In this section we will learn how to remove a toolkit using the ADK command.

1. Type 'orchestrate --help' to list all the orchestrate commands (A)

```
[ADK >
[ADK >orchestrate tools list
[ADK >orchestrate tools remove -n forecast
[INFO] - Successfully removed tool forecast
[ADK >orchestrate --help
```

Name	Title	Description	Tags	API URL	Chat Params	Config	Nickname	App ID	ID
Assistant Agents									
Name	Title	Description	Tags	Nickname	CRN	Instance URL	Assistant ID	Environment ID	ID

Name	Description	Type	Toolkit	App ID
forecast	Weather information for given WGS84 latitude and longitude coordinates.	openapi		

2. Note the toolkits command in the possible orchestrate actions (A)

```
[ADK >orchestrate --help

Usage: orchestrate [OPTIONS] COMMAND [ARGS]...

Options
  --version          Show the installed version of the ADK and Developer Edition Tags
  --debug            Enable debug mode
  --install-completion Install completion for the current shell.
  --show-completion  Show completion for the current shell, to copy it or customize the installation.
  --help            Show this message and exit.

Commands
  env          Add, remove, or select the activate env other commands will interact with (either your local server or a production instance)
  agents       Interact with the agents in your active env
  toolkits     Interact with the toolkits in your active env
  knowledge-bases Upload knowledge your agents can search through to your active env
  connections  Interact with the agents in your active env
  voice-configs Configure voice providers to enable voice interaction with your agents
  server       Manipulate your local Orchestrate Developer Edition server [requires entitlement]
  chat         Launch the chat ui and CLI chat interface for your local Developer Edition server [requires entitlement]
  models       List the available large language models (llms) that can be used in your agent definitions
  channels     Configure channels where your agent can exist on (such as embedded webchat, whatsapp, etc.)
  phone        Manage phone configurations (global resources that can be attached to multiple agents)
  evaluations  Evaluate the performance of your agents in your active env
  settings     Configure the settings for your active env
  partners     Generate a well-structured, submission-ready agent artifact package for partner-built agents
```

3. To learn more about the actions you can perform on a toolkit, type the following command (A) and note the remove option (B).

Orchestrate toolkits --help

```
wxo-ADK --zsh -- 161x39

--show-completion    Show completion for the current shell, to copy it or customize the installation.
--help              Show this message and exit.

Commands
env                Add, remove, or select the activate env other commands will interact with (either your local server or a production instance)
agents             Interact with the agents in your active env
tools              Interact with the tools in your active env
toolkits           Interact with the toolkits in your active env
knowledge-bases    Upload knowledge your agents can search through to your active env
connections        Interact with the agents in your active env
voice-configs      Configure voice providers to enable voice interaction with your agents
server             Manipulate your local Orchestrate Developer Edition server [requires entitlement]
chat               Launch the chat ui and CLI chat interface for your local Developer Edition server [requires entitlement]
models             List the available large language models (llms) that can be used in your agent definitions
channels           Configure channels where your agent can exist on (such as embedded webchat, whatsapp, etc.)
phone              Manage phone configurations (global resources that can be attached to multiple agents)
evaluations        Evaluate the performance of your agents in your active env
settings           Configure the settings for your active env
partners           Generate a well-structured, submission-ready agent artifact package for partner-built agents

[ADK] >Orchestrate toolkits --help  A
Usage: Orchestrate toolkits [OPTIONS] COMMAND [ARGS]...

Interact with the toolkits in your active env

Options
--help    Show this message and exit.

import
add
list
remove
export  B
```

4. You can now learn the precise command to remove a toolkit with the following command (A).

orchestrate toolkits remove --help

```
connections    Interact with the agents in your active env
voice-configs  Configure voice providers to enable voice interaction with your agents
server         Manipulate your local Orchestrate Developer Edition server [requires entitlement]
chat           Launch the chat ui and CLI chat interface for your local Developer Edition server [requires entitlement]
models         List the available large language models (llms) that can be used in your agent definitions
channels       Configure channels where your agent can exist on (such as embedded webchat, whatsapp, etc.)
phone          Manage phone configurations (global resources that can be attached to multiple agents)
evaluations    Evaluate the performance of your agents in your active env
settings       Configure the settings for your active env
partners       Generate a well-structured, submission-ready agent artifact package for partner-built agents

[ADK] >Orchestrate toolkits --help
Usage: Orchestrate toolkits [OPTIONS] COMMAND [ARGS]...

Interact with the toolkits in your active env

Options
--help    Show this message and exit.

Commands
import
add
list
remove
export

[ADK] >orchestrate toolkits remove --help  A
Usage: orchestrate toolkits remove [OPTIONS]

Options
* --name -n TEXT Name of the toolkit you wish to remove [required]
--help    Show this message and exit.
```

To remove a toolkit, a name must be specified. The toolkit removal command are:

```
orchestrate toolkits remove --name <toolkit name>
```

or

```
orchestrate toolkits remove -n <toolkit name>
```

You can use this command from the ADK to remove all the toolkits you may have created import MCP tools for examples.

Summary

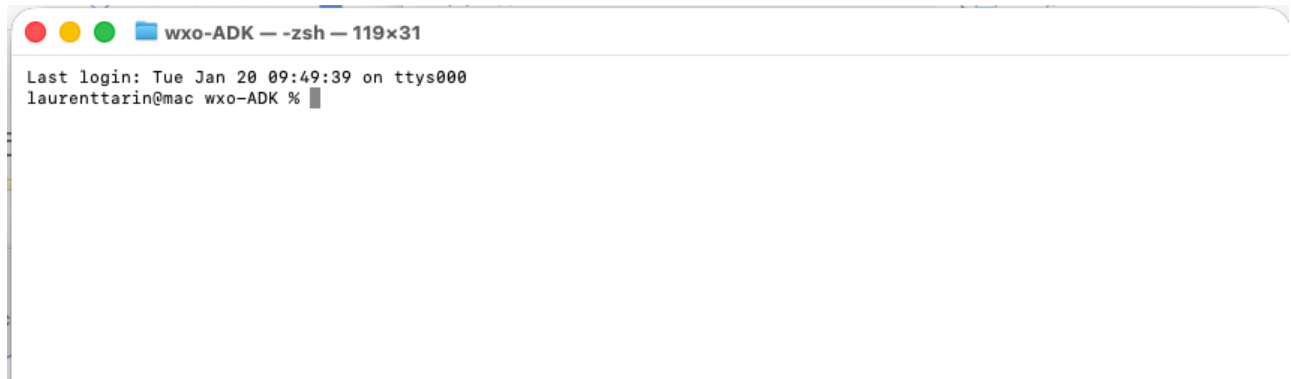
During this lab you have learn the basics of using the ADK to work with a SaaS environment. You have learned how to swap from the local ADK environment you may have already used in a previous ADK lab, to a SaaS environment and control this SaaS environment from the ADK.

You have also learned how easy it is to navigate and learn the different orchestrate ADK commands using the inline help.

Appendix: Reset your environment

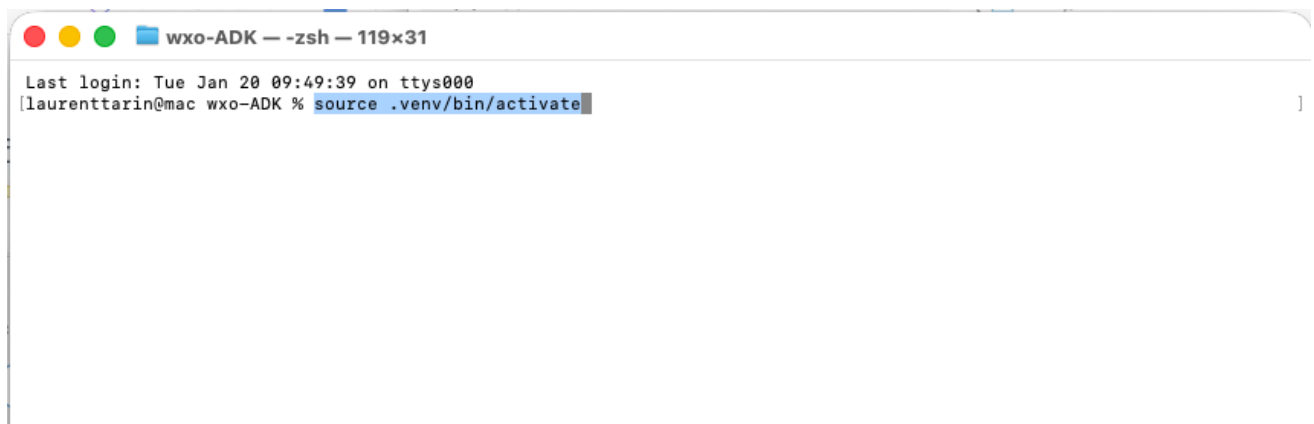
These steps will remove any existing tools and agents from your environment and restart the local watsonx Orchestrate server in readiness to start the lab.

1. Open a Terminal window on your **wxO-ADK** folder (A).



2. Enter the following command into the terminal to start your virtual environment:

```
source .venv/bin/activate
```



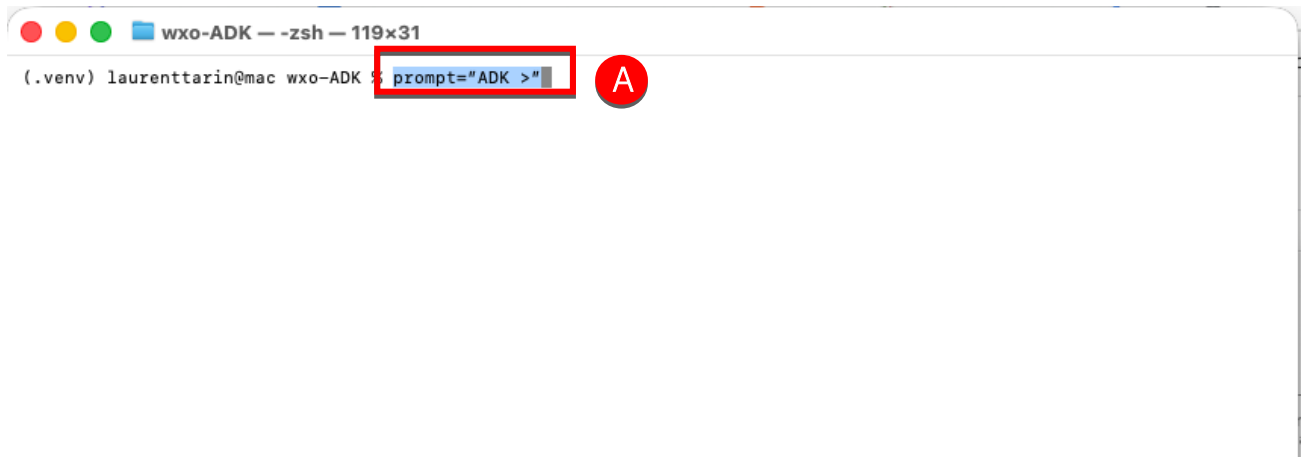
3. Check that the prompt starts with (.venv) (A) indicating that you are in your virtual environment.



A terminal window titled "wxo-ADK - zsh - 119x31". The output shows the last login time and the execution of the command `source .venv/bin/activate`. The prompt has changed from `laurenttarin@mac wxo-ADK %` to `(.venv) laurenttarin@mac wxo-ADK %`, indicating the virtual environment is active.

4. Enter the following command into the terminal to improve the command visibility, then press Enter:

```
MS Windows users: function prompt{"ADK>"}  
Mac users: prompt="ADK >"
```



A terminal window titled "wxo-ADK - zsh - 119x31". The prompt is `(.venv) laurenttarin@mac wxo-ADK %`. The command `prompt="ADK >"` is being entered and is highlighted with a red box. A red circle with the letter 'A' is next to the command, indicating it is the step to be performed.

5. Enter the following command into the terminal to stop your chat service, then press Enter.

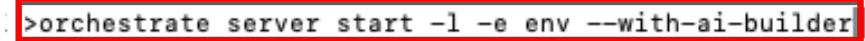
```
orchestrate chat stop
```

6. Enter the following command into the terminal to reset your server, then press Enter.

```
orchestrate server reset
```

7. Enter the following command into the terminal to start the server with your env configuration, then press Enter.

```
orchestrate server start -l -e env --with-ai-builder
```

wxo-ADK — -zsh — 161x39

```
>orchestrate server start -l -e env --with-ai-builder
```



Note: Some services may fail to start on the first attempt, and you may encounter issues connecting to the docker repository. Retry the start command in this situation, depending on the volume of requests, it can take several attempts to connect.

When startup is complete the terminal output will resemble the example below, restart can take 2-3 minutes to complete. The local server will be ready when you will see this **INFO** in the terminal screen (A):

```
wxo-ADK -- -zsh -- 119x31

Skipping already applied migration: 20251001.sql
Skipping already applied migration: 20251006.sql
Skipping already applied migration: 20251007.sql
Skipping already applied migration: 20251008.sql
Skipping already applied migration: 20251009.sql
Skipping already applied migration: 20251022.sql
Skipping already applied migration: 20251023.sql
Skipping already applied migration: 20251029.sql
Skipping already applied migration: 20251101.sql
Skipping already applied migration: 20251103.sql
Skipping already applied migration: 20251104.sql
Skipping already applied migration: 20251107.sql
Skipping already applied migration: 20251110.sql
Skipping already applied migration: 20251111.sql
Skipping already applied migration: 20251112.sql
Existing wxo_observability DB found
Skipping already applied observability migration: 00_init_db.sql
Skipping already applied observability migration: 01_create_tables.sql
Skipping already applied observability migration: 02_triggers_and_functions.sql
Skipping already applied observability migration: 03_security.sql
Skipping already applied observability migration: 04_rls_policies.sql
Skipping already applied observability migration: 20250116.sql
[INFO] - Migration ran successfully.
[INFO] - Waiting for orchestrate server to be fully initialized and ready...
[INFO] - Orchestrate services initialized successfully
[INFO] - local tenant found
[INFO] - You can run 'orchestrate env activate local' to set your environment or 'orchestrate chat start' to start the
UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3010, username: orchestrate@ibm.com, pa
ssword: orchestrate
ADK
```



You are now ready to work with watsonx Orchestrate ADK.

